

Intel – and its role in the semiconductor industry

Dezső Sima

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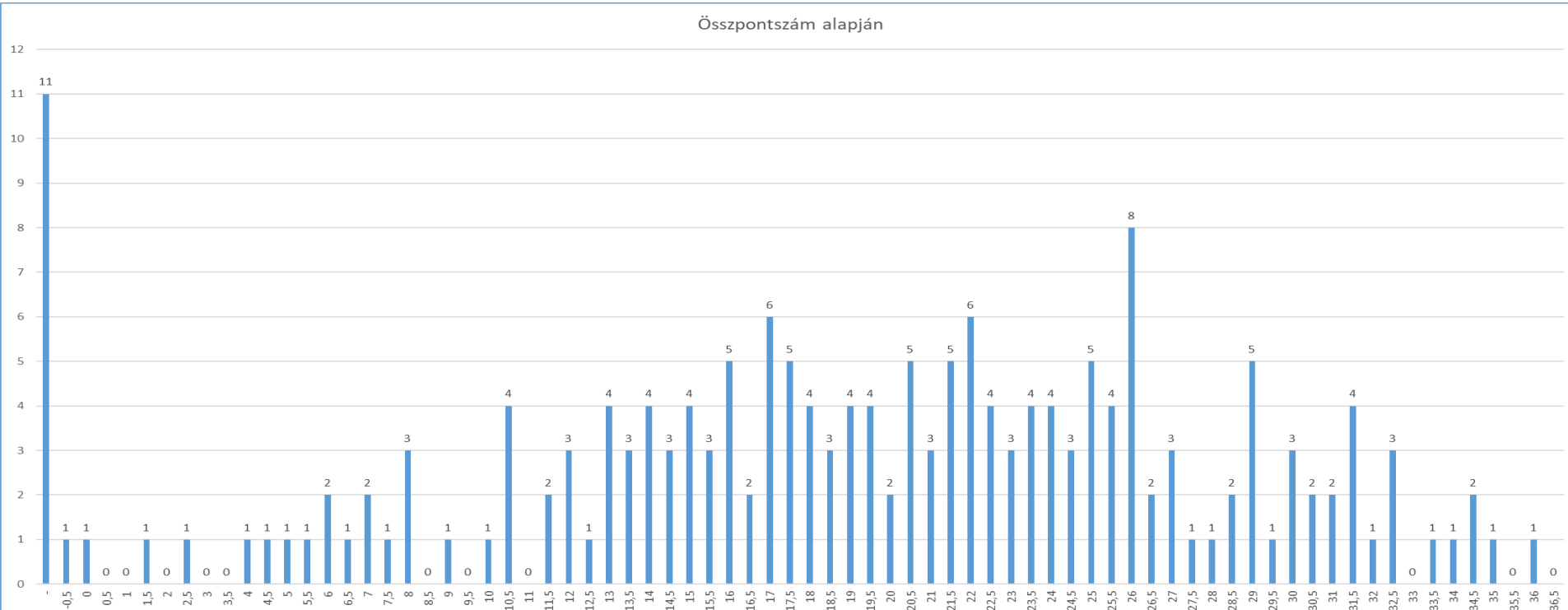
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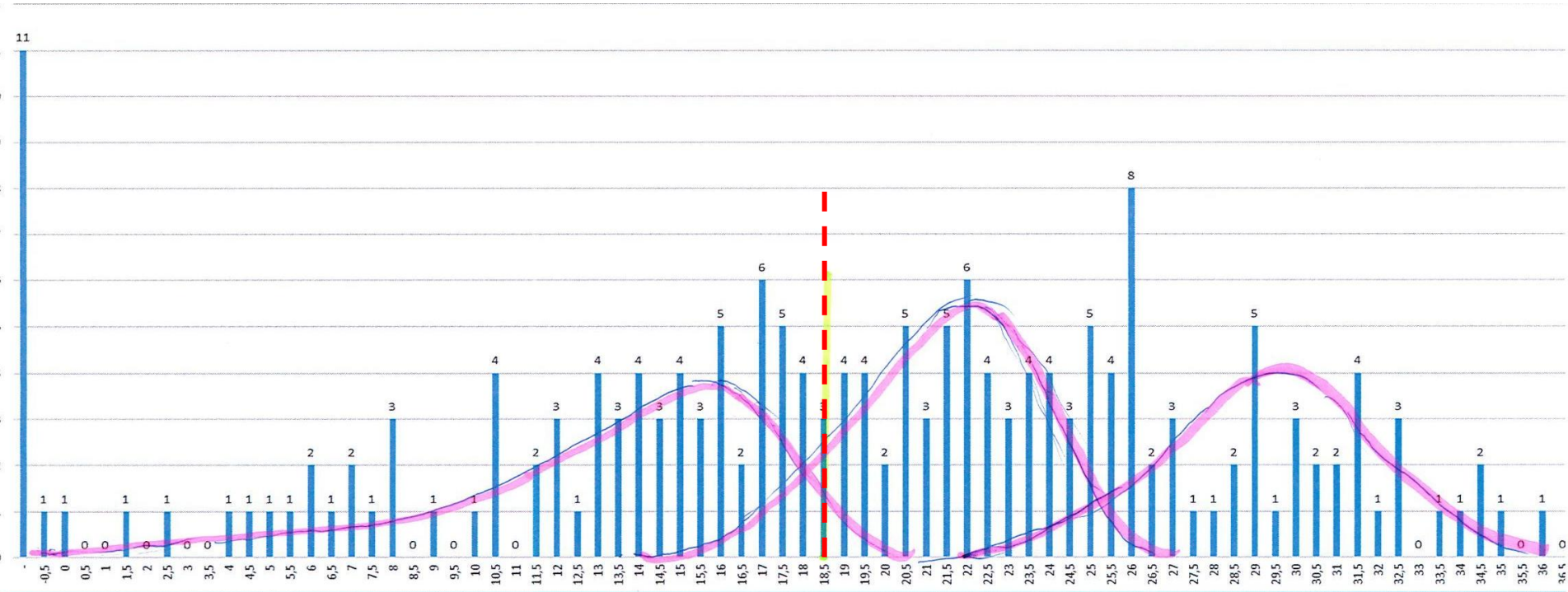
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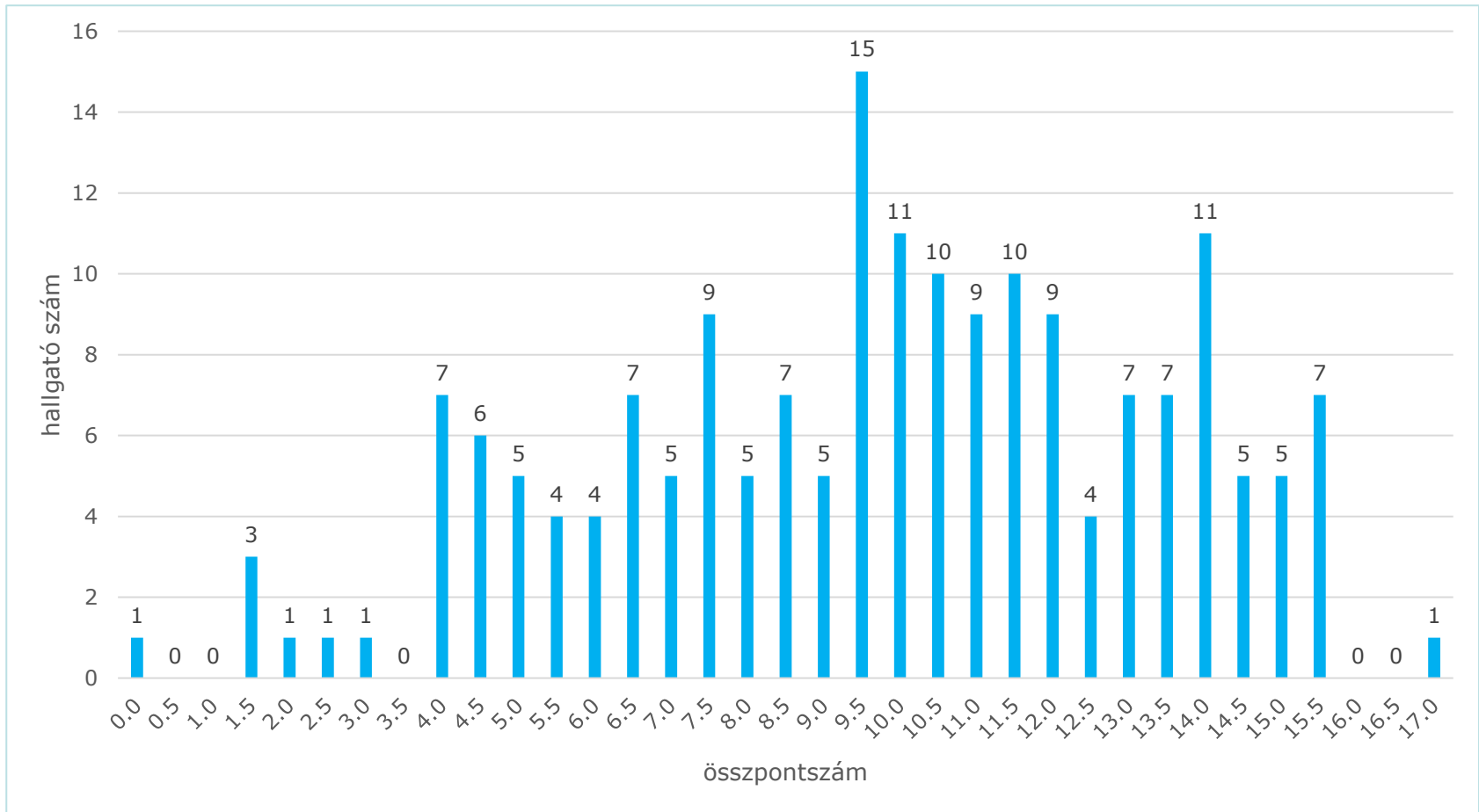


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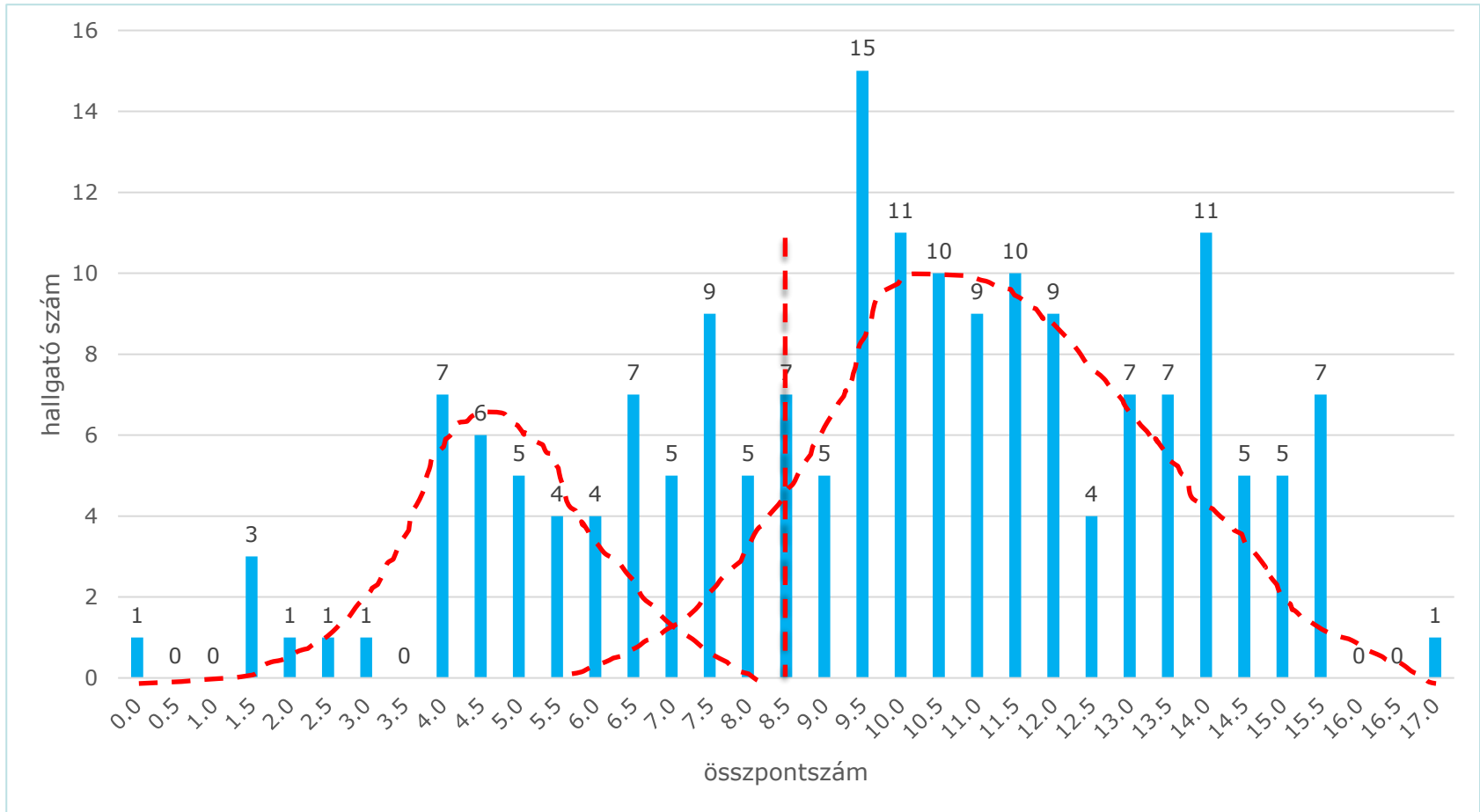
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Korszerű számítógép architektúrák ZH 2025. 10. 29 Hallgatói összpontszámok eloszlása



Korszerű számítógép architektúrák ZH 2025. 10. 29 Hallgatói összpontszámok eloszlása



Intel – and its role in the semiconductor industry

- 1. How did Intel become dominant in the processor market?
- 2. The race for dominance in the processor market
- 3. What is Intel's current market position?
- 4. References

Intel – and its role in the semiconductor industry

- This Chapter is devoted to Intel.
- We will trace its journey **from inception to its rise** as the world's leading semiconductor company.
- Next, we provide an overview of the **fierce competition** for leadership in IC technology, processor performance, and processor efficiency.
- Finally, we assess **Intel's current position** in the global processor market.

1. How did Intel become dominant in the processor market?

1. How did Intel become dominant in the processor market? (1)

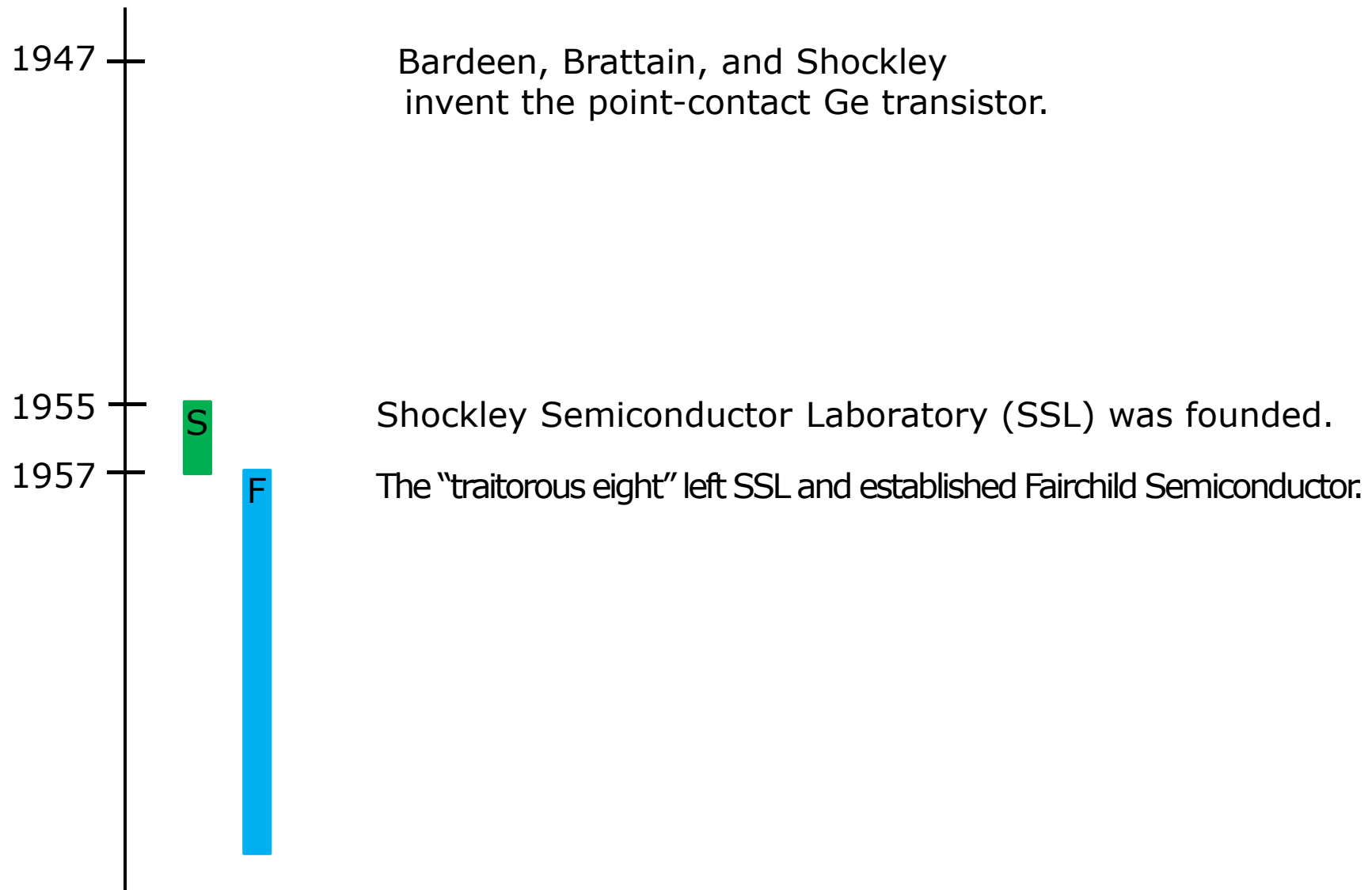
1. How did Intel become dominant in the processor market?

The long road to Intel's founding and its rise to dominate the processor market

Below, is a **timeline**, highlighting major events leading to the firm's establishment.

1. How did Intel become dominant in the processor market? (2)

The background: The emergence of the semiconductor industry



1. The foundation of Intel and its rise to dominate the processor market (3)

The "Traitorous Eight," who went on to establish Fairchild Semiconductor

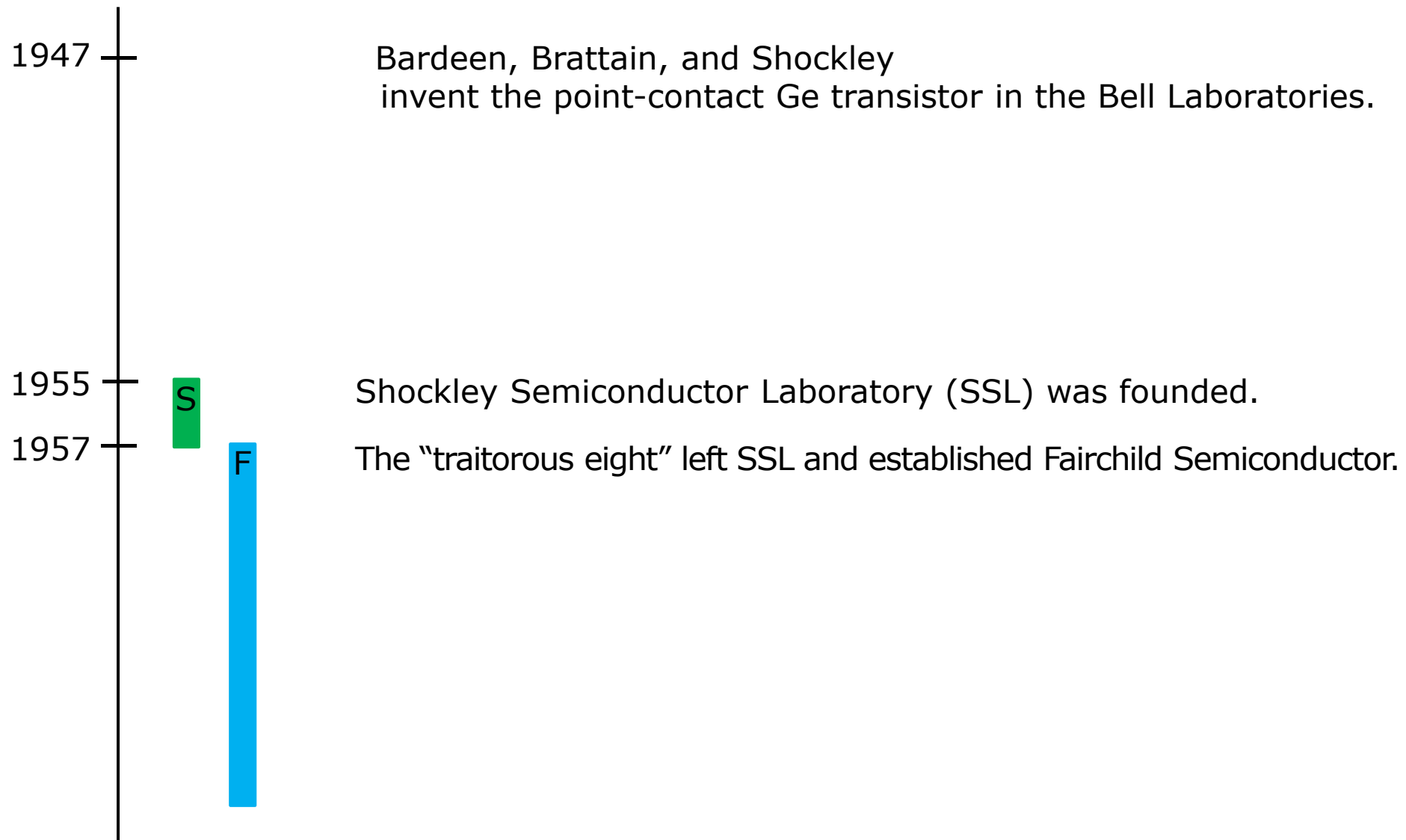


Picture: The "traitorous eight" [159]

- In 1957, frustration with William Shockley's authoritarian leadership at Shockley Semiconductor Laboratory reached a boiling point: eight of its most talented researchers—feeling undervalued —resigned en masse.
- Determined to build a more collaborative and innovative environment, these "Traitorous Eight" founded Fairchild Semiconductor [159].

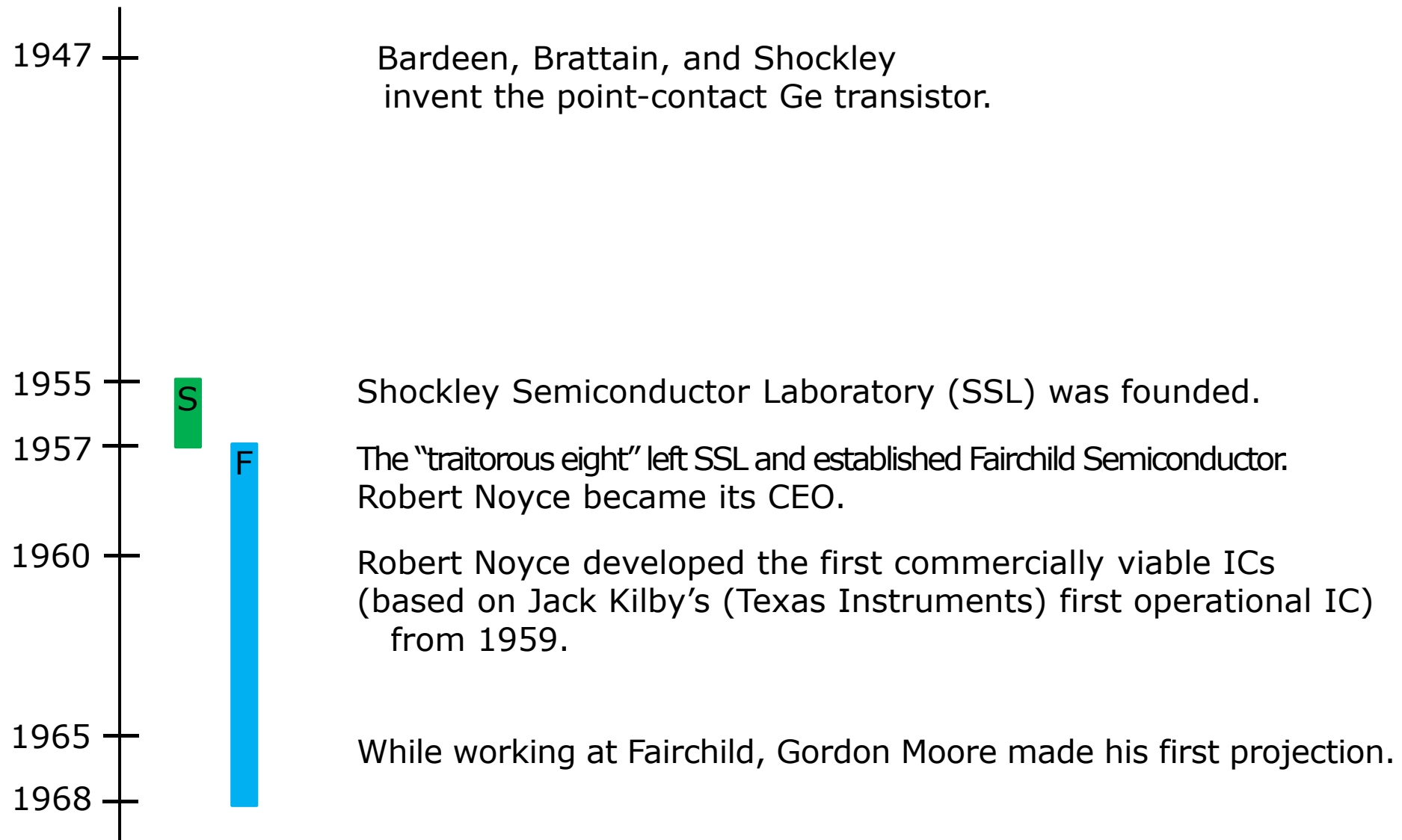
1. The foundation of Intel and its rise to dominate the processor market (4)

The background: The emergence of the semiconductor industry



1. The foundation of Intel and its rise to dominate the processor market (5)

The background: the evolution of semiconductors up to the founding of Intel



1. The foundation of Intel and its rise to dominate the processor market (6)

Gordon Moore's original projection for the growth rate of the number of transistors in integrated circuits (ICs) from 1965

- In 1965, Gordon Moore examined the growth rate in transistor counts on integrated circuits since their introduction in 1959.
- He noted that the number of transistors per chip roughly doubled each year. He published these findings in a seminal paper, and this observation later became known as **Moore's original law**.

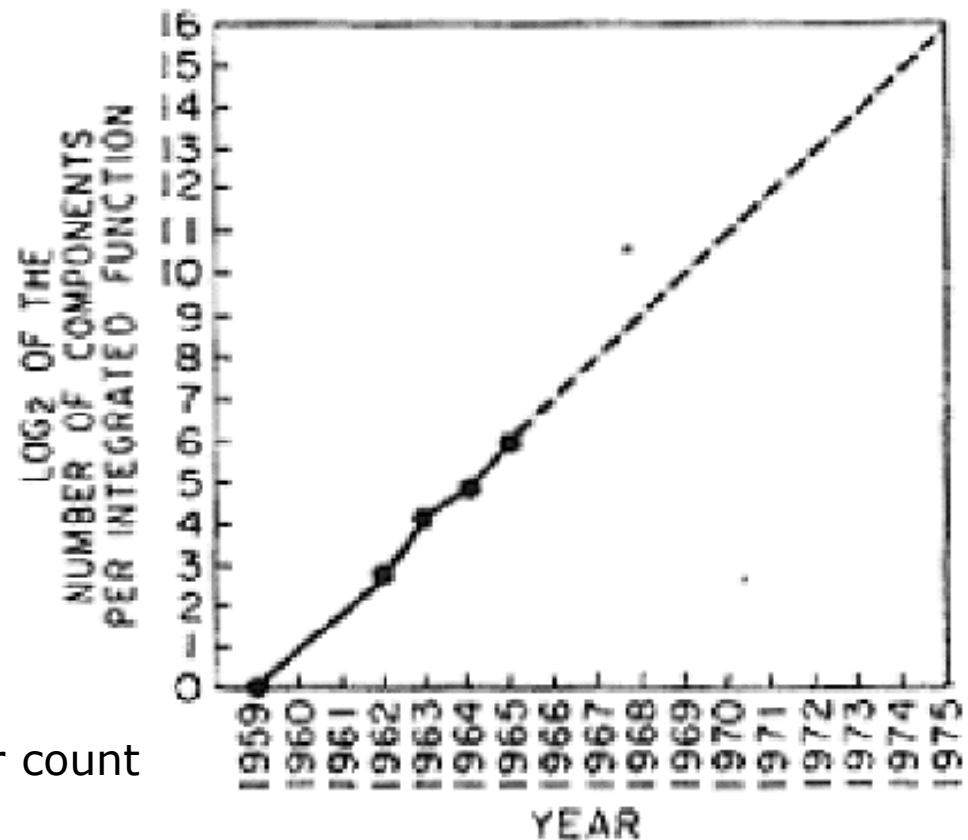


Figure: Moore's original projection about the growth rate in transistor count on integrated circuits [9]

1. The foundation of Intel and its rise to dominate the processor market (7)

Moore's revised projection for raising transistor counts/die from 1975 [10]

In 1975, Gordon Moore revisited his 1965 projection and acknowledged that the evolution of IC technology had slowed down, and made the forecast that starting around 1980, the transistor counts/die would double roughly every two years. as seen below.

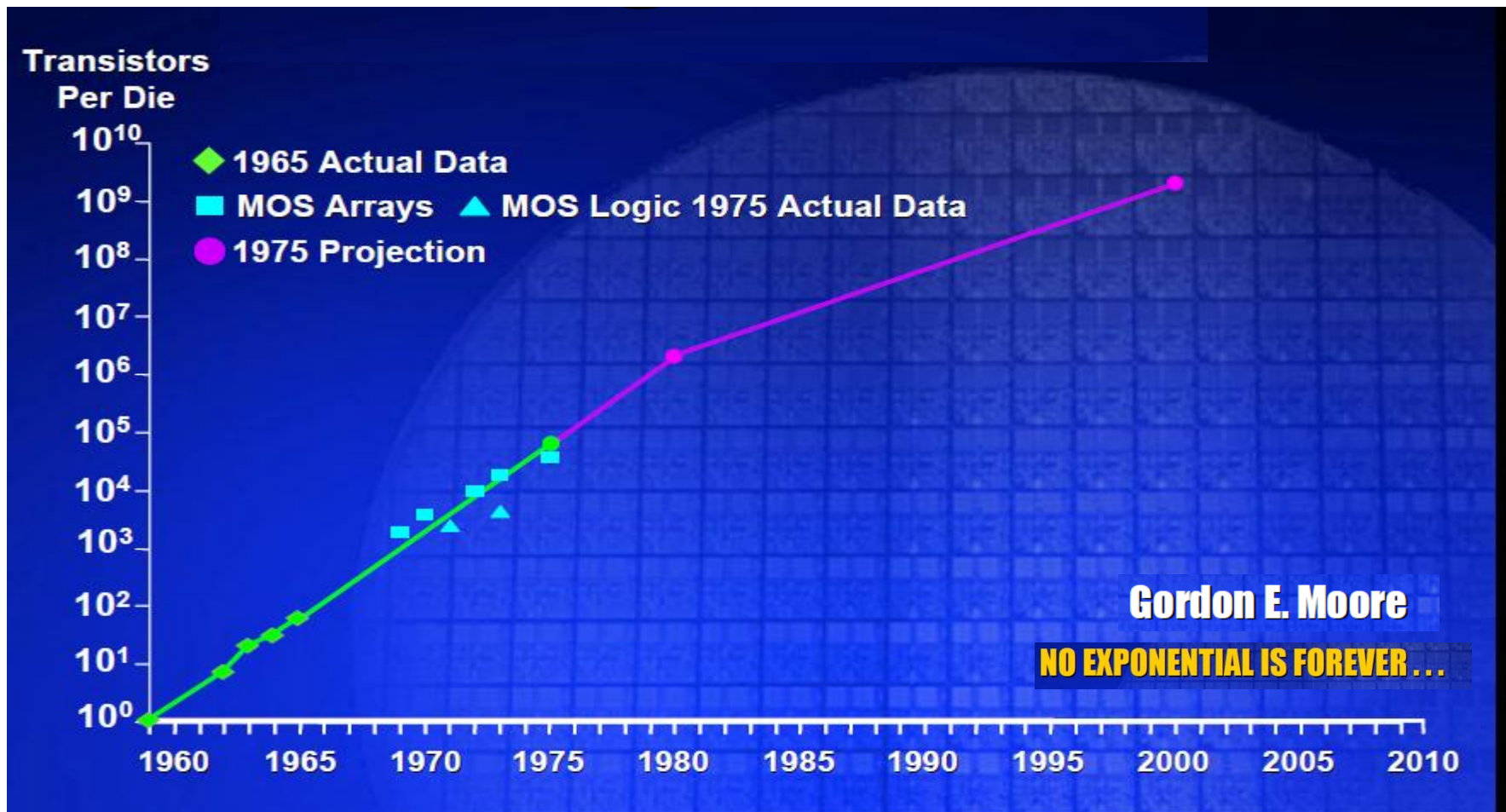
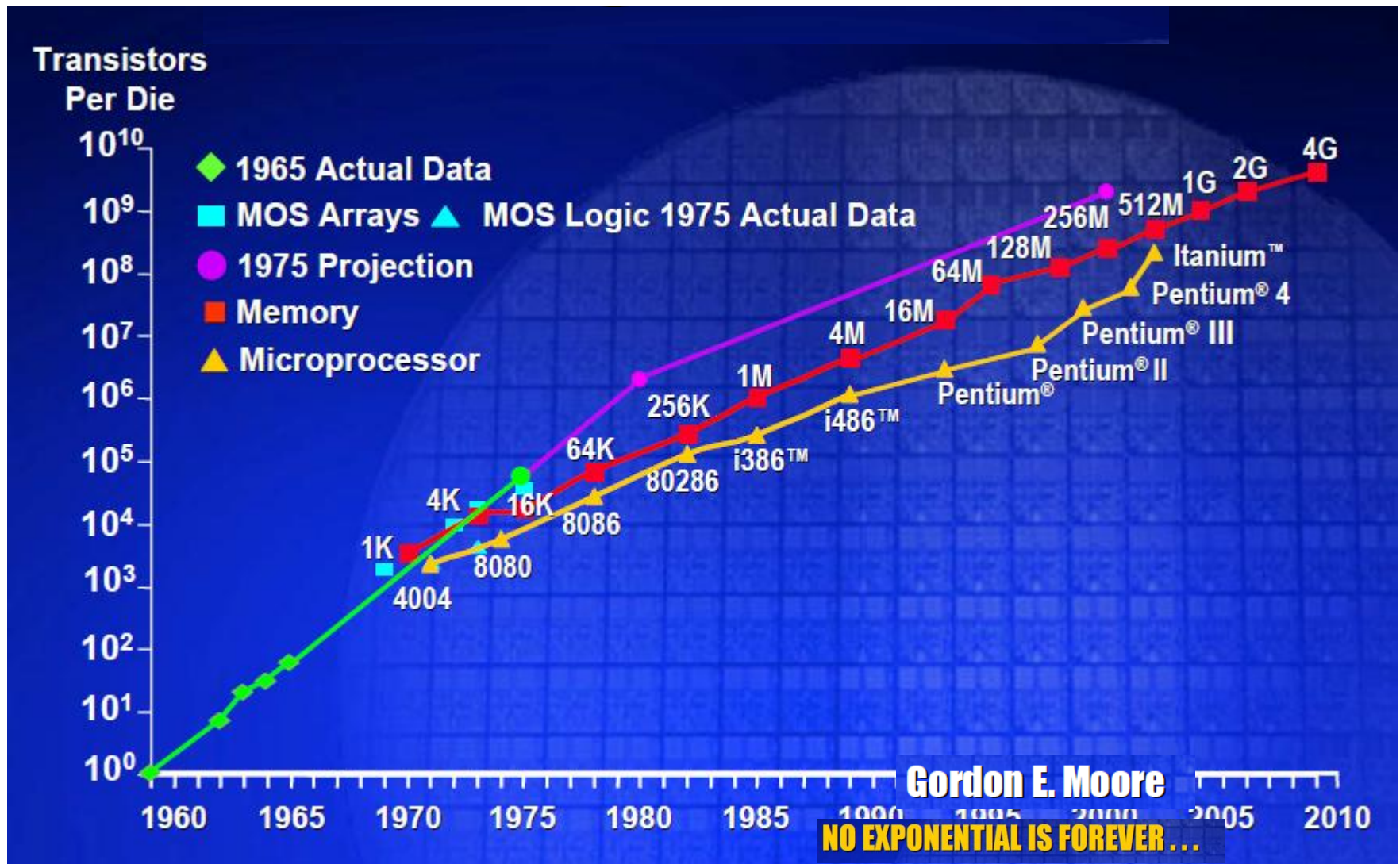


Figure: Moore's revised projection from 1975.

1. The foundation of Intel and its rise to dominate the processor market (8)

Moore's affirmation of its 1975 projection from 2003 [10]



Actual data from 2003 confirm (more or less) Moore's updated projection.

1. The foundation of Intel and its rise to dominate the processor market (9)

Confirming Moore's revised law after the 2010 interval [11]

(Presented at AMD's Financial Analyst Day on May 16, 2017)

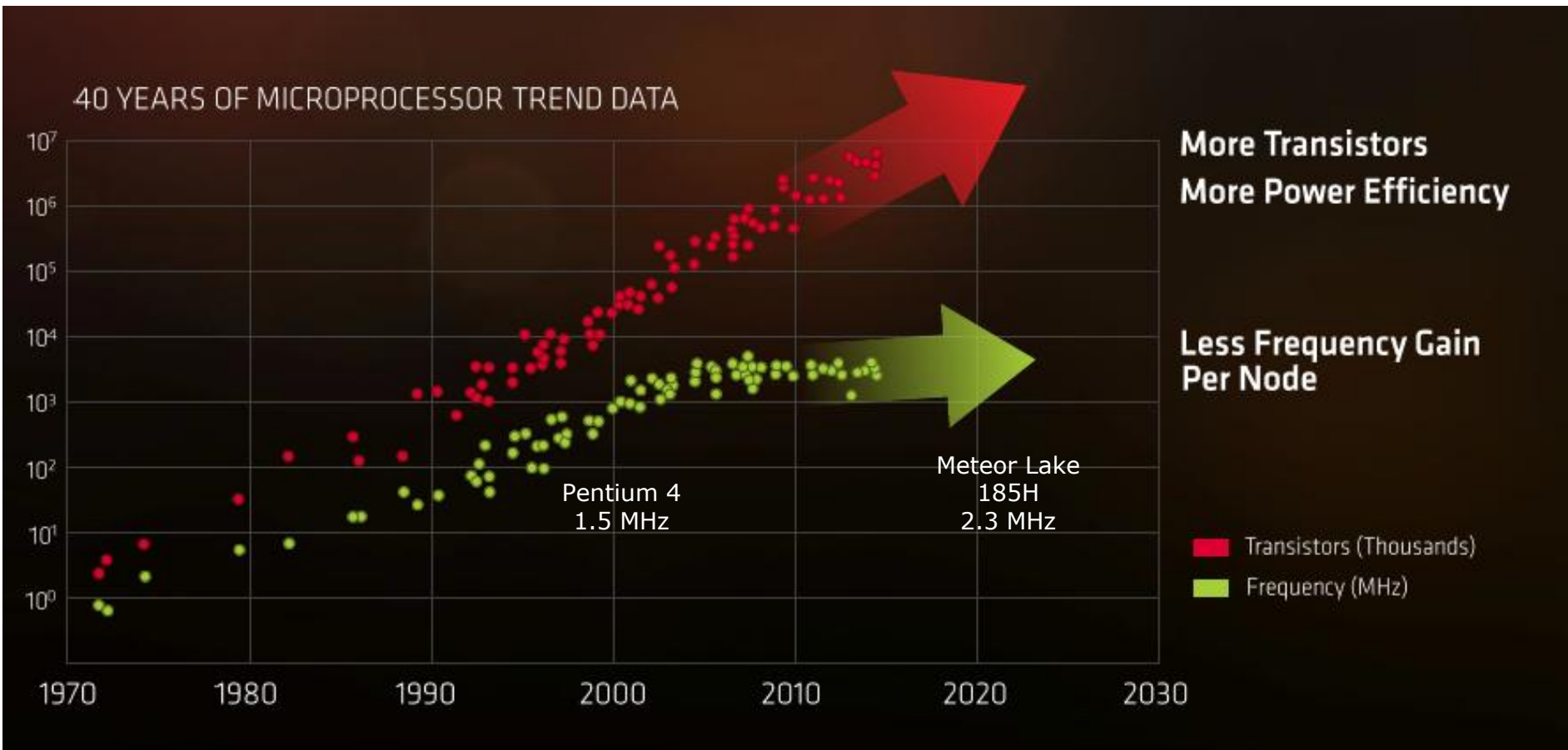


Figure: Rising the transistor counts in line with Moore's Law but stagnating CPU clock speeds around the early 2000s — hovering at roughly 3–4 GHz—as thermal and power-density limits halted further frequency gains.

1. The foundation of Intel and its rise to dominate the processor market (10)

Mismanagement at Fairchild leading to the foundation of a number of semiconductor companies

- However, despite its technological breakthroughs, Fairchild's progress was hindered by poor management decisions made by its parent organization.
- Many scientists at Fairchild Semiconductor became frustrated with the **poor management decisions of the parent company**, as well as the lack of share bonuses.
- This dissatisfaction catalyzed a **wave of departures**, as several key employees left the company towards the end of the 1960s, to establish their own semiconductor firms, as shown in the following table.

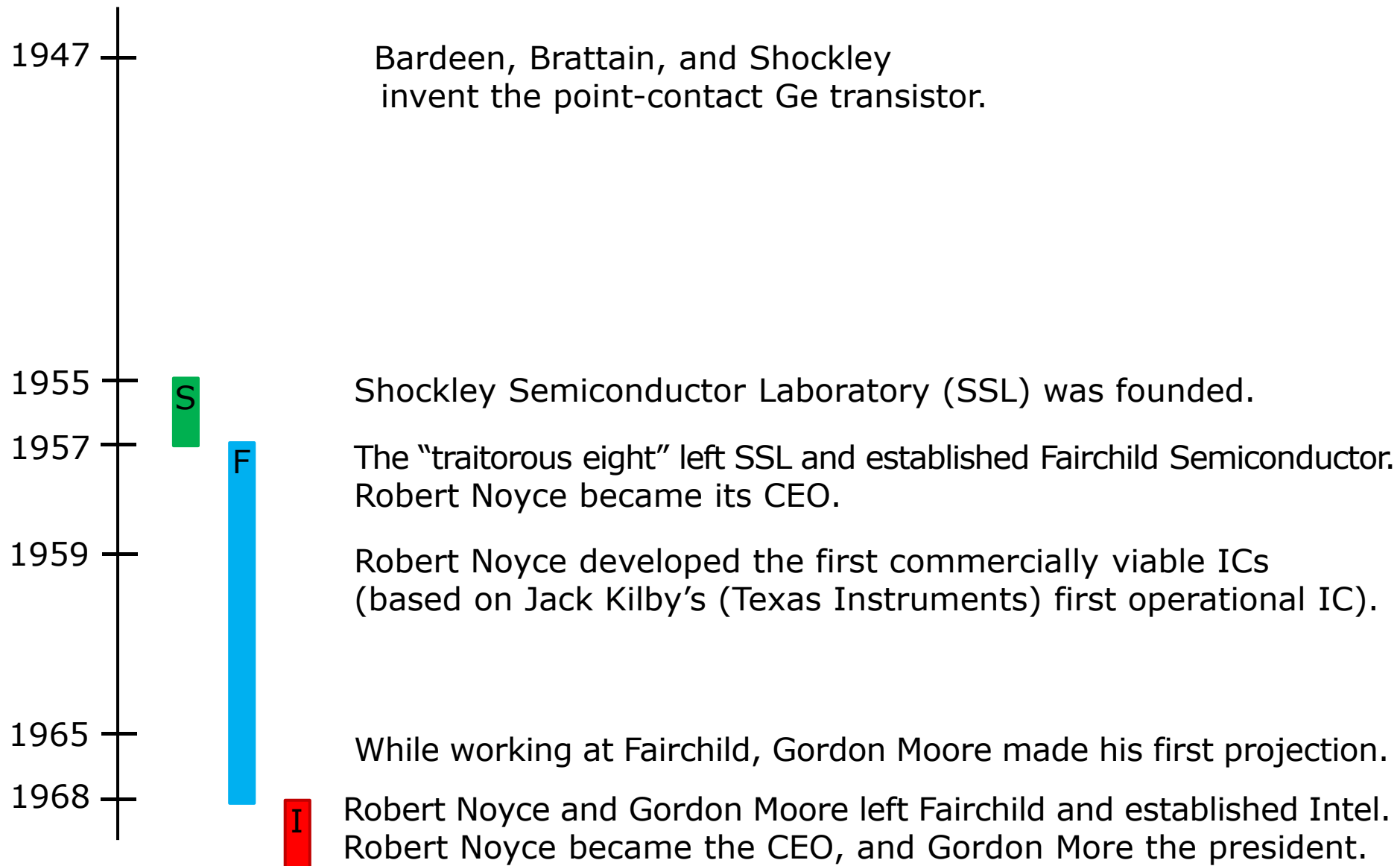
Company	Year Founded	Year Acquired	Acquirer
AMD	1968		
Avantek	1968	1991	HP
Intel	1967		
Intersil	1967	1988	Harris Semi
Linear Technology	1981	2017	Analog Devices
LSI Logic	1981	2014	Avago
National Semi	1967	2011	Texas Instruments
Signetics	1961	1975	Philips (Now NXP)
Synertek	1973	1979	Honeywell

Table: Companies founded by former employees of Fairchild, called the Fairchildren [1]

- From all the companies founded by former Fairchild employees, **only AMD and Intel survived**.

1. The foundation of Intel and its rise to dominate the processor market (11)

The foundation of Intel by Robert Noyce and Gordon Moore in 1968



1. The foundation of Intel and its rise to dominate the processor market (12)

Robert Noyce and Gordon Moore



Robert Noyce (1927-1990) [160]



Gordon Moore (1936-2023) [161]

1. The foundation of Intel and its rise to dominate the processor market (13)

Intel's third and fourth employees

Intel's third and fourth employees were **Andy Grove** (Gróf András) and **Leslie Vadász** (Vadász László).

Both individuals left Hungary in 1956 at the age of 20.

1. The foundation of Intel and its rise to dominate the processor market (14)

Andy Grove (Gróf András) (1936-2016)

- He served as Intel's **third CEO** (Chief Executive Officer) **from 1987 to 1998**. During his leadership, Intel rose to become the world's **largest semiconductor company**.
- In recognition of his extraordinary achievements, Time Magazine named him "**Man of the Year**" in **1997**.



1997 marked the **50th anniversary** of the **invention of the transistor**.

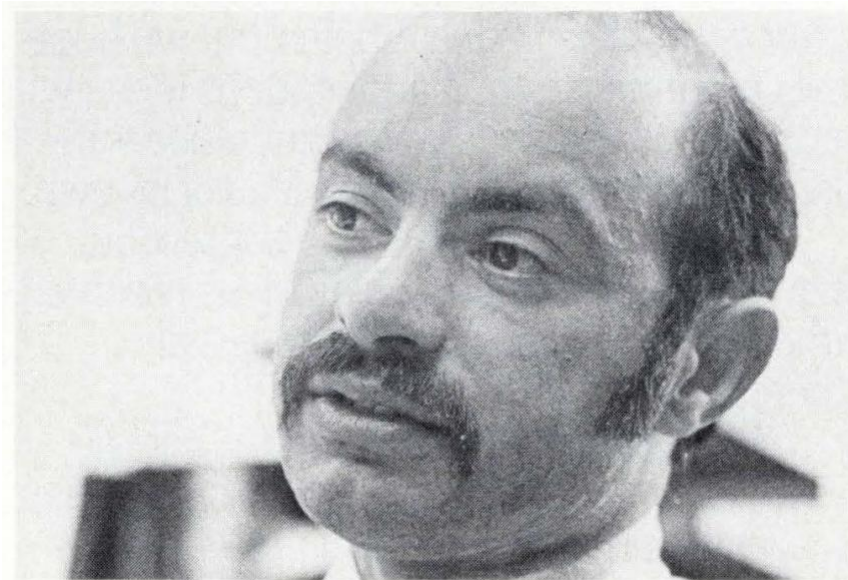
Choosing him as Man of the Year, rather than Shockley, Brattain, Bardeen, Moore, Noyce, etc., was an extraordinary acknowledgment of his achievements at Intel's top.

Andy Grove, the Man of the Year in 1997 [136]

1. The foundation of Intel and its rise to dominate the processor market (15)

Leslie Vadasz (Vadász László) (1936-)

- He led the team that developed the Intel 1103 **1 Kb DRAM memory**, the world's **first commercially available memory chip**.
- This marked a turning point in computing history; semiconductor (**DRAM**) **memory** soon replaced **magnetic core memory** as the industry standard.

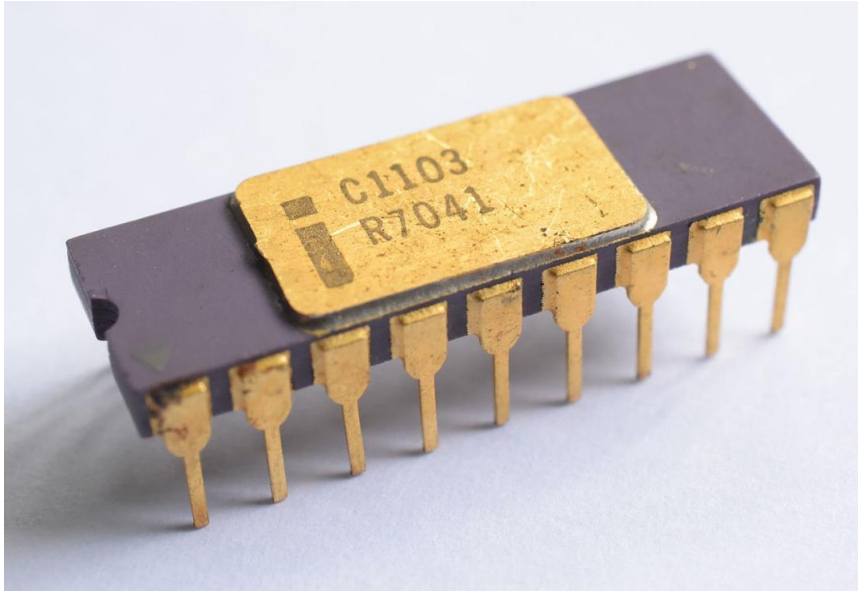


Happy Hungarian. Leslie Vadasz, who was head of the 1103 design team, now manages all Intel engineering.

Leslie Vadasz [127]

1. The foundation of Intel and its rise to dominate the processor market (16)

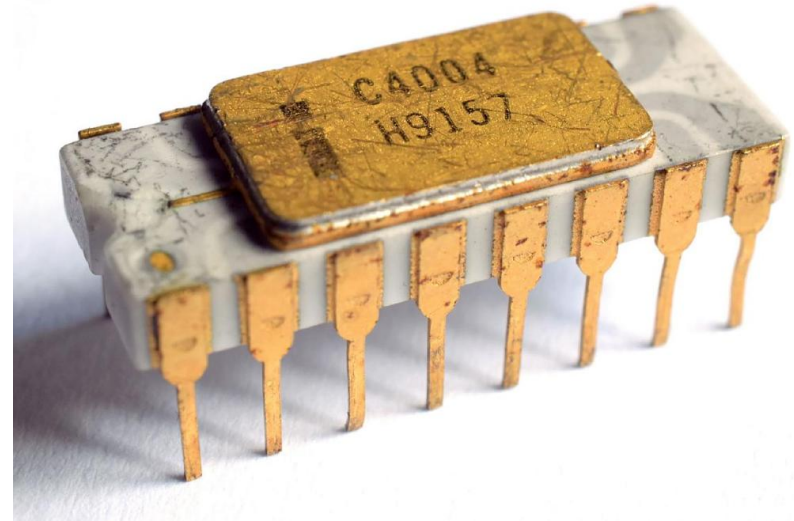
Intel's first commercially available chips: the 1103 DRAM and the 4004 4-bit microprocessor chips



Picture: The 1103 DRAM,
released in 10/1970.

It includes more than 3000 transistors.
[155]

It was the **first successful DRAM chip**,
it replaced bulky magnetic memory.



Picture: The 4004 microprocessor,
actually, a CPU,
released in 11/1971.

It includes 2300 transistors. [156]

It marks **the world's first microprocessor**,
a groundbreaking innovation that laid
to modern computing.

1. How did Intel become dominant in the processor market? (16)

Prediction for the emergence of commercially viable ICs, such as 1 Kb DRAMs or 4-bit microprocessors based on Moore's projection

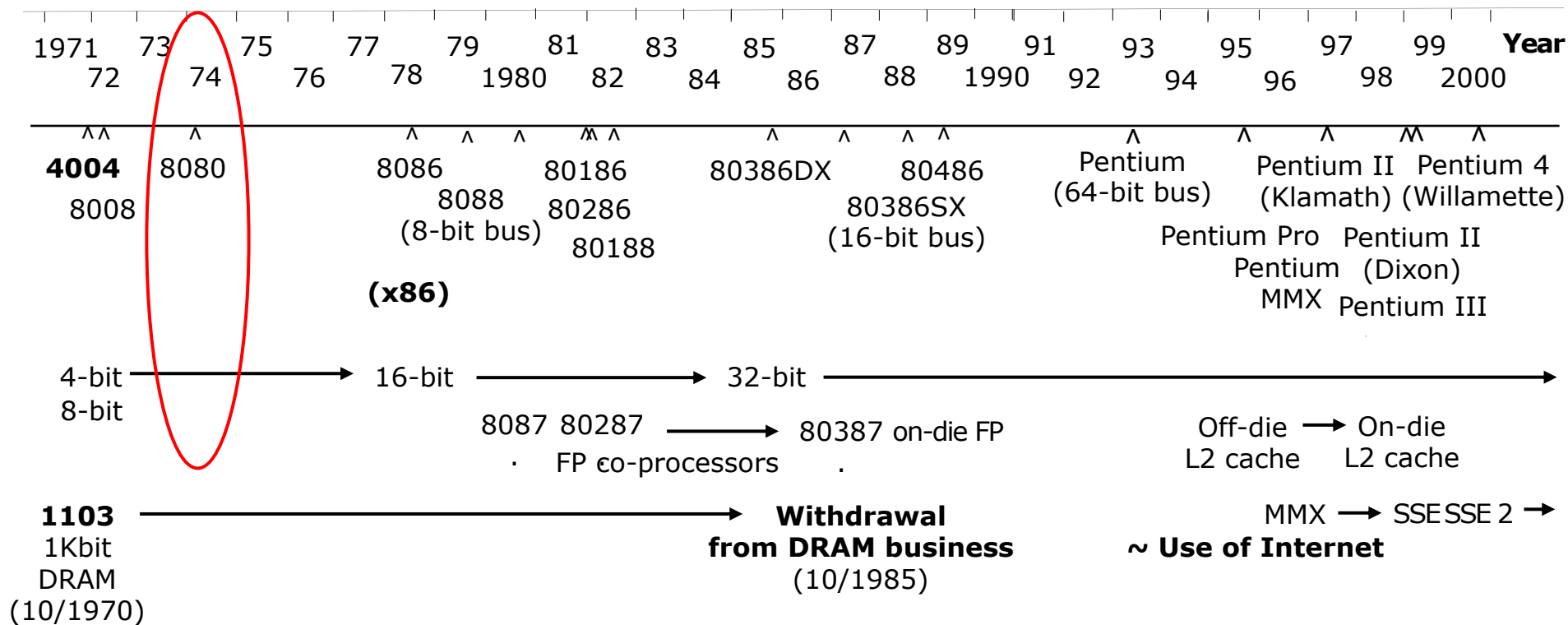
Commercially viable 1 Kb DRAM memories and 4-bit microprocessors require several thousand transistors, as indicated previously.

According to Moore's original projection, beginning from 1959, it would take about ten years to reach the transistor counts necessary for these devices.

It is therefore not surprising that Intel introduced the first 1 Kb DRAMs in 1970 and the 4-bit microprocessors in 1971.

1. How did Intel become dominant in the processor market? (17)

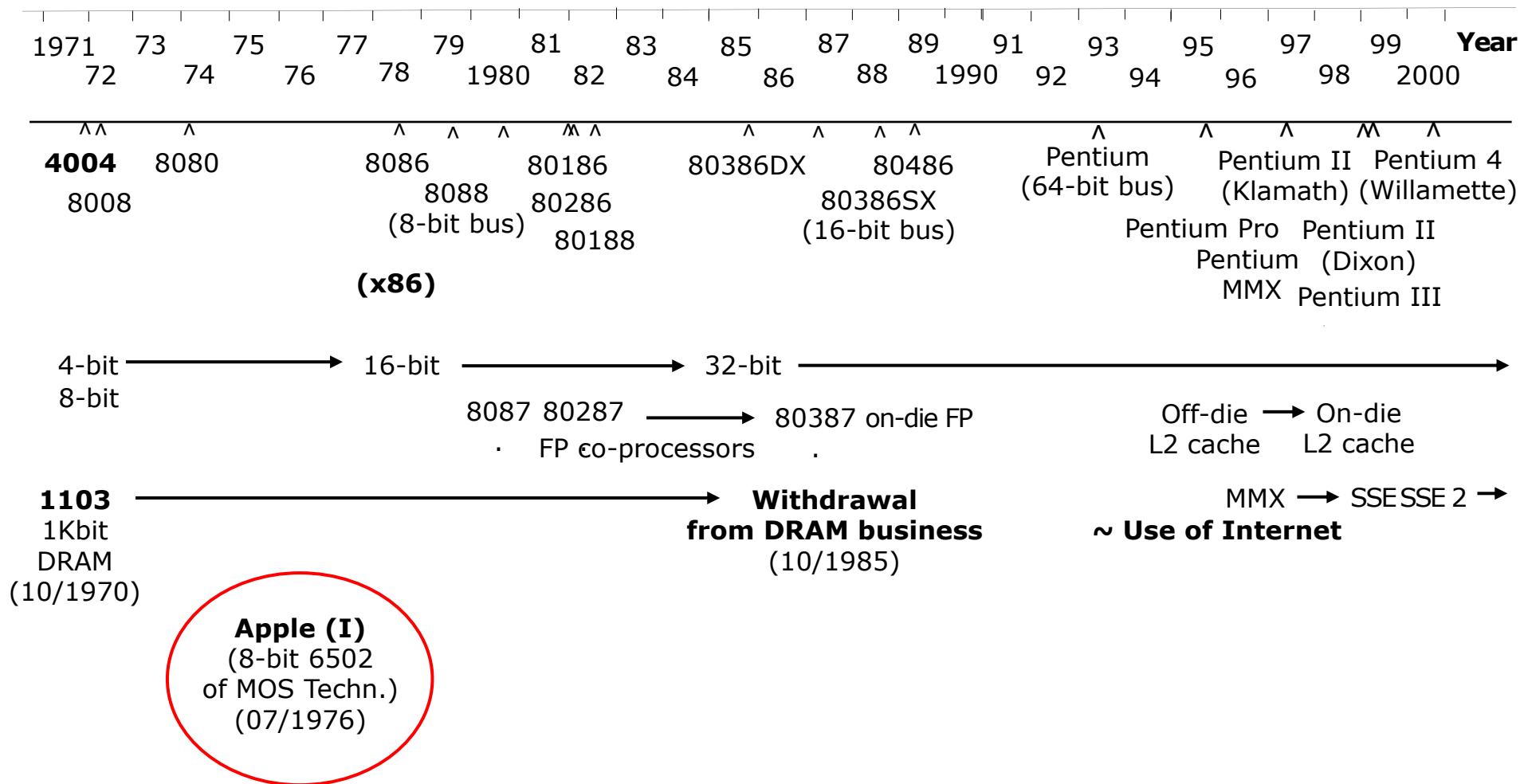
Intel's early mainstream processors and DRAM memories -1



Following the release of the 4-bit 4004, and the 8-bit 8008, Intel's first widely adopted microprocessor was the 8080.

1. How did Intel become dominant in the processor market? (18)

Apple's decision to opt for the 6502 for its Apple (I) computer and its consequences -1



1. How did Intel become dominant in the processor market? (19)

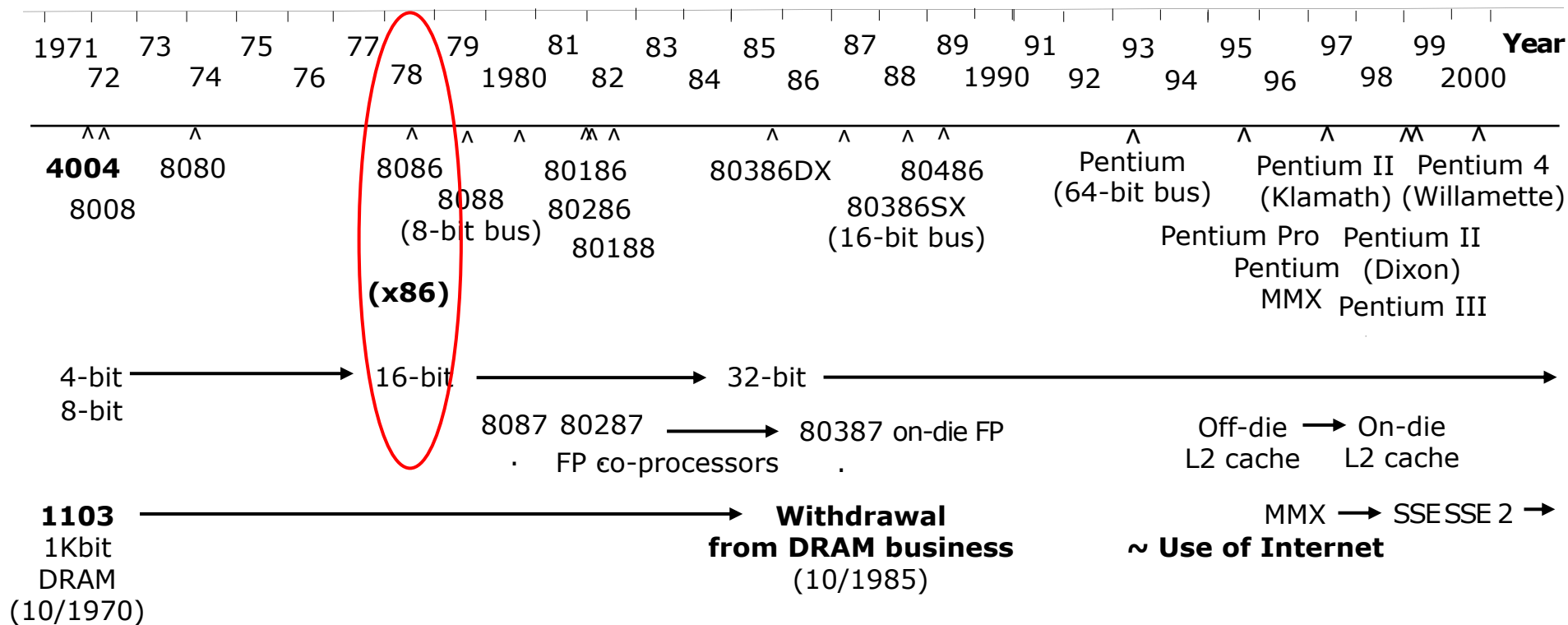
Consequences of Apple's decision to opt for the 6502 for its Apple (I) computer -2

It should be noted that for its Apple (I) PC the company selected MOS Technology's 8-bit 6502 processor rather than Intel's 8080 because the 6502 was considerably cheaper.

This decision had far-reaching consequences, leading to a divergence between Apple's and Intel's processor lines.

1. How did Intel become dominant in the processor market? (19b)

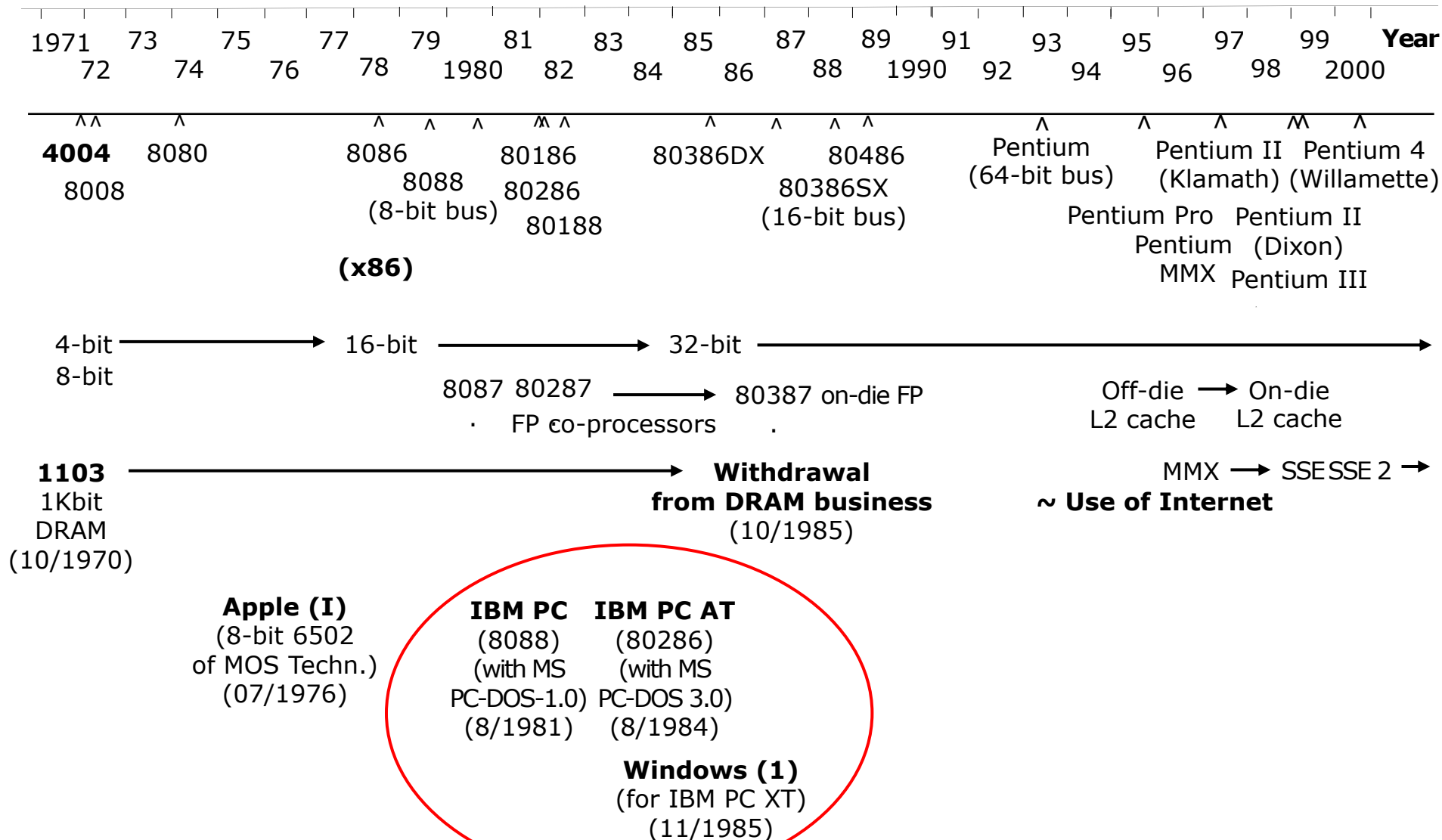
Intel's early mainstream processors and DRAM memories -1



- Following the 8-bit 8080, the 16-bit 8086 processors were launched in 1978.
- The 8086 plays a crucial role in computing history by establishing the x86 ISA, which became the cornerstone of Intel's mainstream processors.

1. How did Intel become dominant in the processor market? (20)

IBM's far-reaching decisions about which processor and operating system should be used for its IBM PC in 1981 -1



1. How did Intel become dominant in the processor market? (21)

IBM's far-reaching decisions about which processor and operating system should be used for its IBM PC in 1981 -2

- Further **significant decisions** were made **by IBM** that had a lasting impact on the computer industry.
- First, **for its IBM PC**, the company opted for **Intel's 16-bit 80188** processor, which had an 8-bit wide bus to help reduce costs.
- Subsequently, the **IBM PC** became the **market leader** and inspired other vendors **to clone** this device.

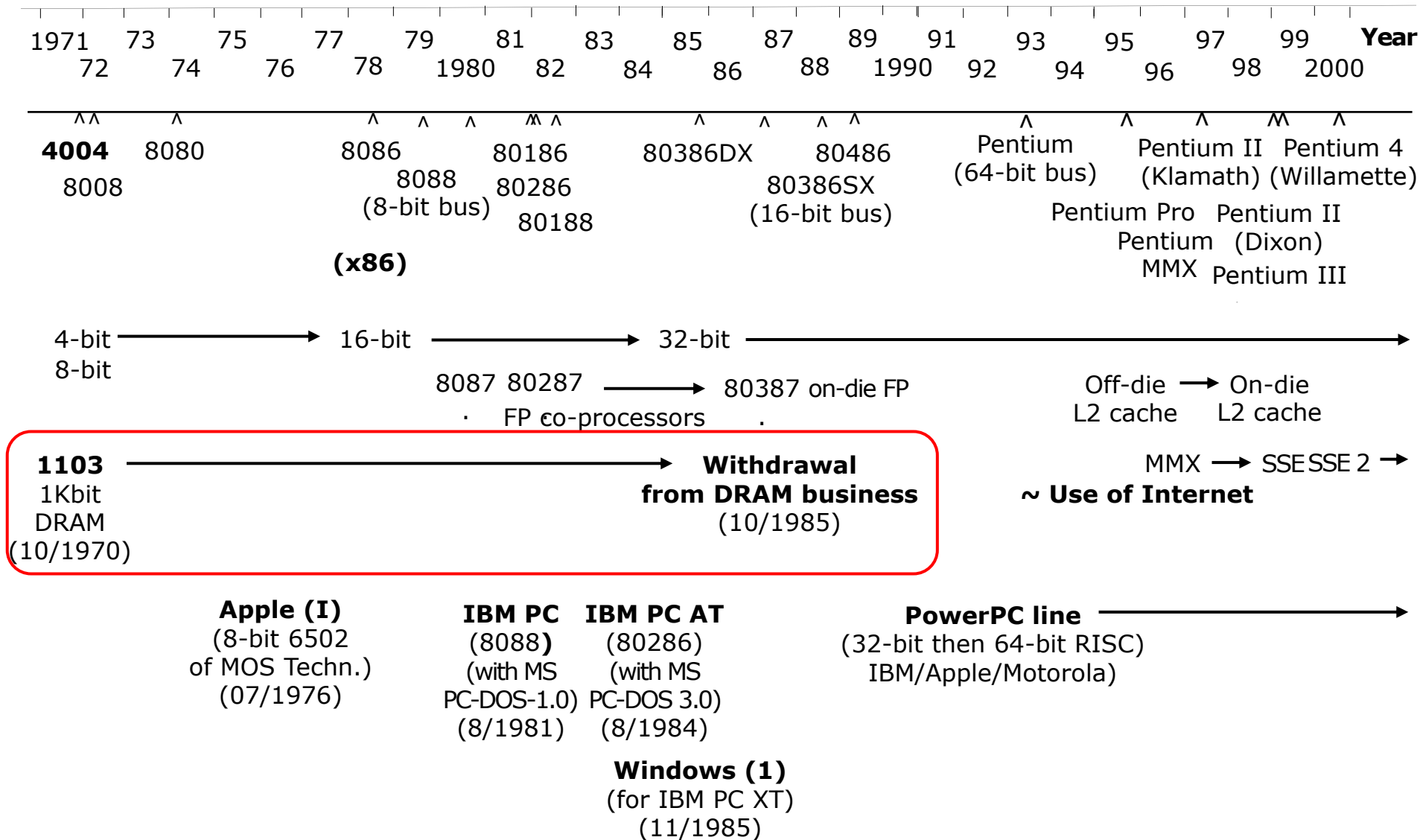
This **largely contributed to Intel's rise** to become the dominant provider of microprocessors worldwide.

- Another important decision made by IBM was **to ask Microsoft to provide an operating system** for its IBM PC.

This partnership enabled **Microsoft to become the leading provider of x86-based client computers** and to develop the **Windows series of OSs**, which ultimately become dominant in the x86 client market.

1. How did Intel become dominant in the processor market? (22)

Intel's withdrawal from the memory business in 1985 -1



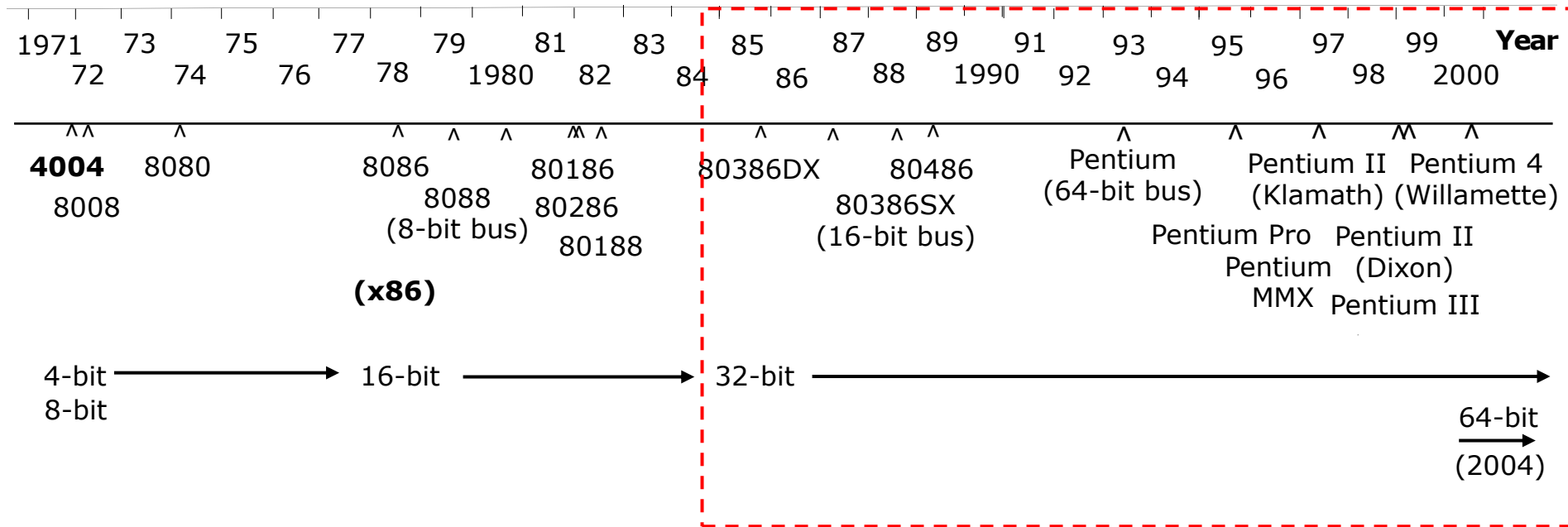
1. How did Intel become dominant in the processor market? (23)

Intel's withdrawal from the memory business in 1985 -2

- In its **early years**, Intel was primarily a **memory company** with significant revenues from DDR memory sales.
- However, in the early 1980s, **Japanese companies**, such as NEC, Toshiba, Hitachi and Fujitsu **entered the memory market** by offering high-quality DRAM products **at dumping prices**.
- This led to a **major shakeup, by mid-1980, several notable US companies**, including Intel, National Semiconductor, AMD, and Mostek had **excited the DRAM business entirely**.

1. How did Intel become dominant in the processor market? (24)

Extending the word length first to 32-bit and then to 64 bit -1



1. How did Intel become dominant in the processor market? (25)

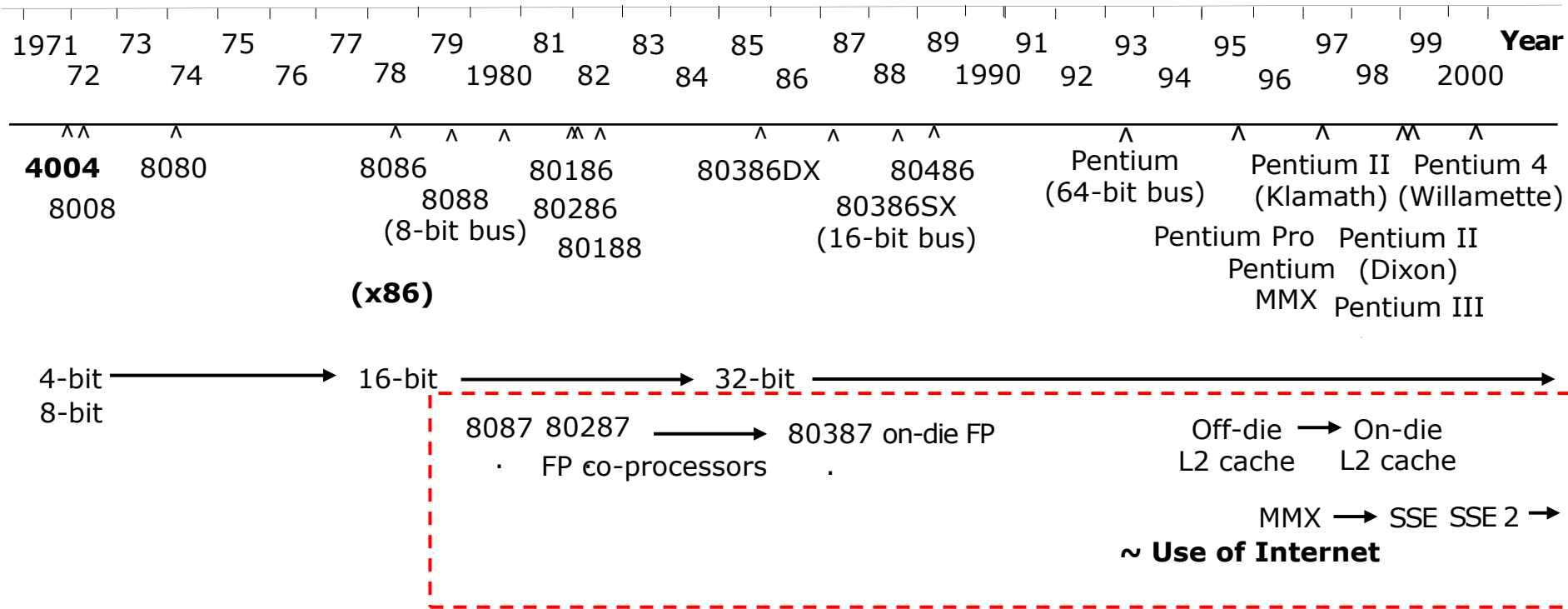
Extending the word length first to 32-bit and then to 64 bit -2

From the previous summary of the evolution of Intel's early processors, the following facts should be highlighted:

- Intel's first 32-bit processor, the 386DX, was released in 1985.
- In 2004, the 3rd core of the Pentium 4 line introduced 64-bit processing.

1. How did Intel become dominant in the processor market? (25b)

Supporting FP processing, multimedia, 3D graphics, and introducing L2 caches -1



For a detailed explanation, see the next slide.

1. How did Intel become dominant in the processor market? (25c)

Supporting FP processing, multimedia, 3D graphics, and introducing L2 caches -2

From the previous summary of the evolution of Intel's early processors, the following facts should be highlighted:

- **FP processing** was initially handled by **coprocessors**, such as the 8087 (1982) to 80387 (1987) co-processors.
- This changed in **1989** with the release of the **80486**, which **integrated FP processing** directly onto the processor die.
- An **off-die L2 cache first** appeared in the **Pentium Pro** processor in **1985**, and was later **integrated** onto the die with the Celeron's **Mendocino core** in **1998**.
- Meanwhile, in the **mid-1990s**, the **internet** became widely adopted, reshaping the computing landscape.

To support **multimedia** processing, in **1996**, Intel introduced its **64-bit FX SIMD processing** technology, known as **MMX**, with the **Pentium MMX** processor.

This was followed in **1999** by the introduction of **128-bit FP SIMD** processing, called **SSE** (Streaming SIMD Extension, in the **Pentium III** to support **3D graphics**.

Intel expanded SIMD capabilities further in **2000** with **SSE2** in the **Pentium 4**, enabling **128-bit SIMD processing for both FX and FP** operations.

- Subsequent generations introduced even wider SIMD extensions, culminating in **256-bit (AVX)** and **512-bit (AVX-512)** **SIMD** instructions for **both FX and FP** operations.

1. How did Intel become dominant in the processor market? (26)

Intel's dominance in the microprocessor market in the 1990s [172]

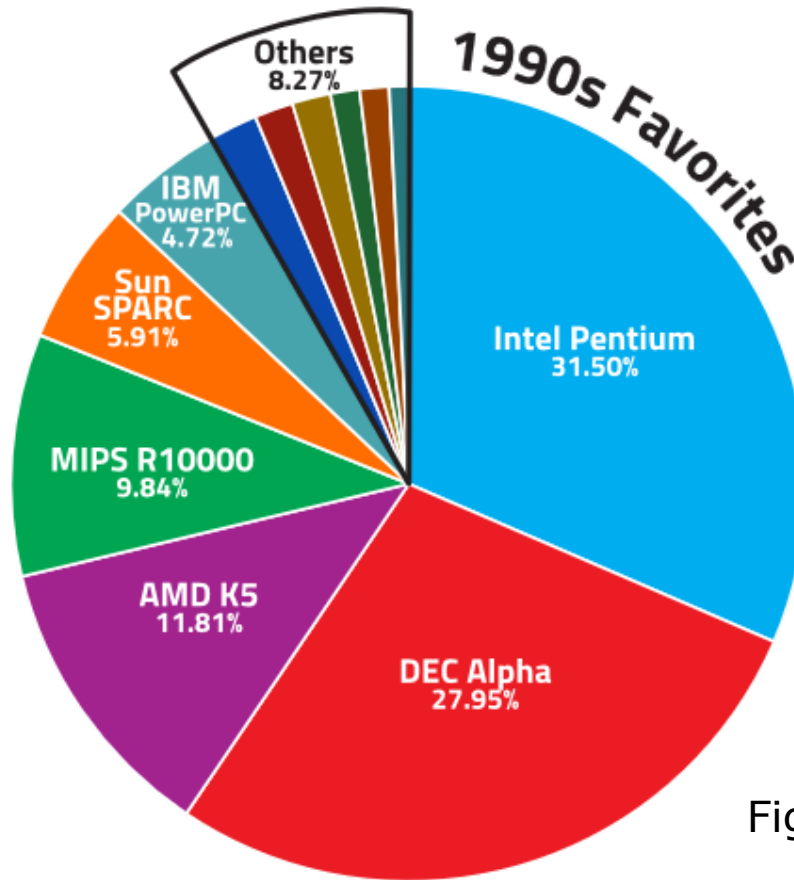


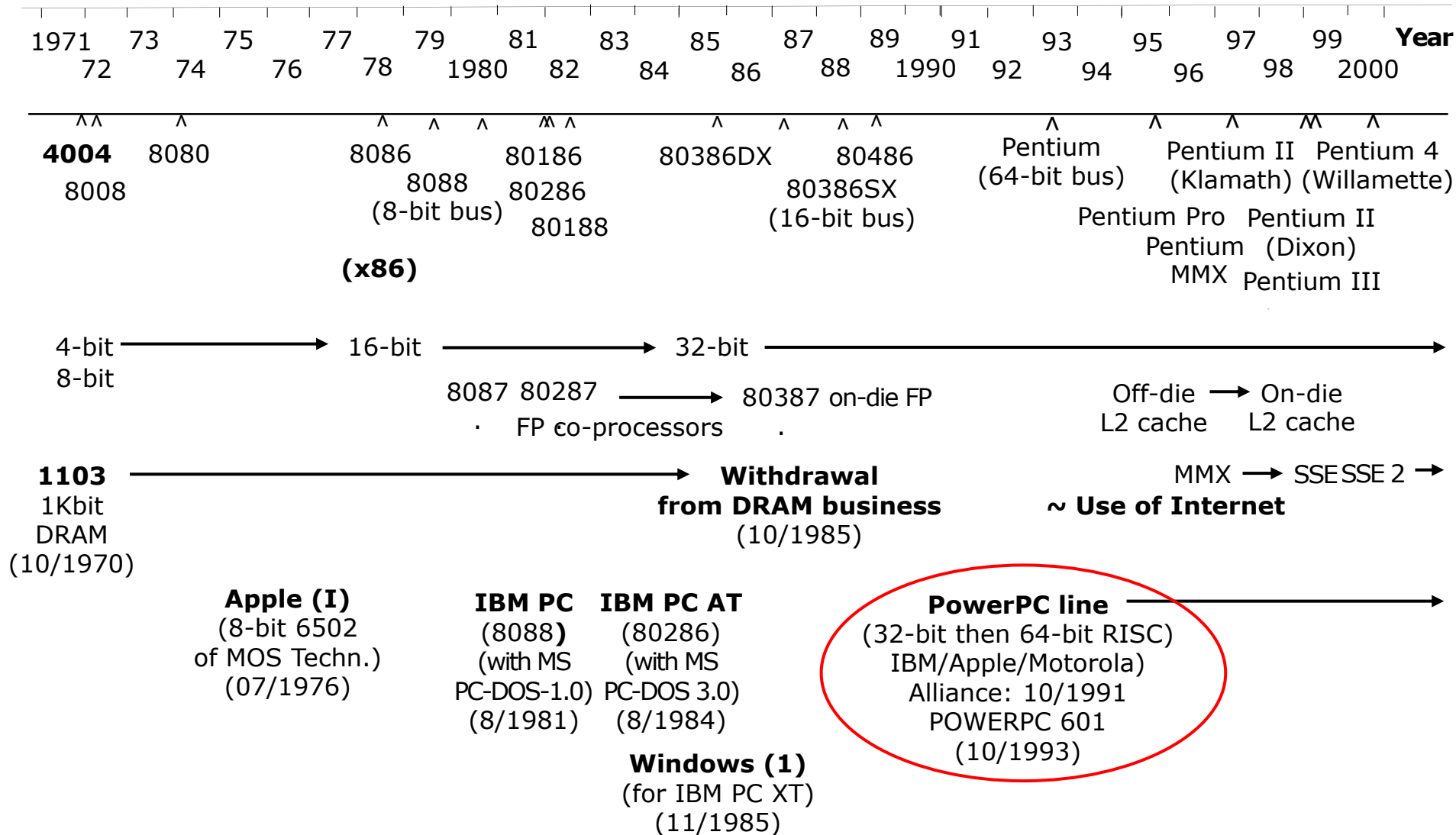
Figure: Microprocessor market share in the 1990s [172]

- The figure illustrates how far Intel dominated the microprocessor market during the 1990s.
- Later, by the mid-2000s, nearly all competing processor families had been discontinued, leaving only Intel and AMD as active players.

This consolidation further solidified Intel's leadership, making its dominance in the industry even more pronounced during this period.

1. How did Intel become dominant in the processor market? (27)

IBM's, Apple's, and Motorola's 1991 alliance to challenge Intel's x86 supremacy -1



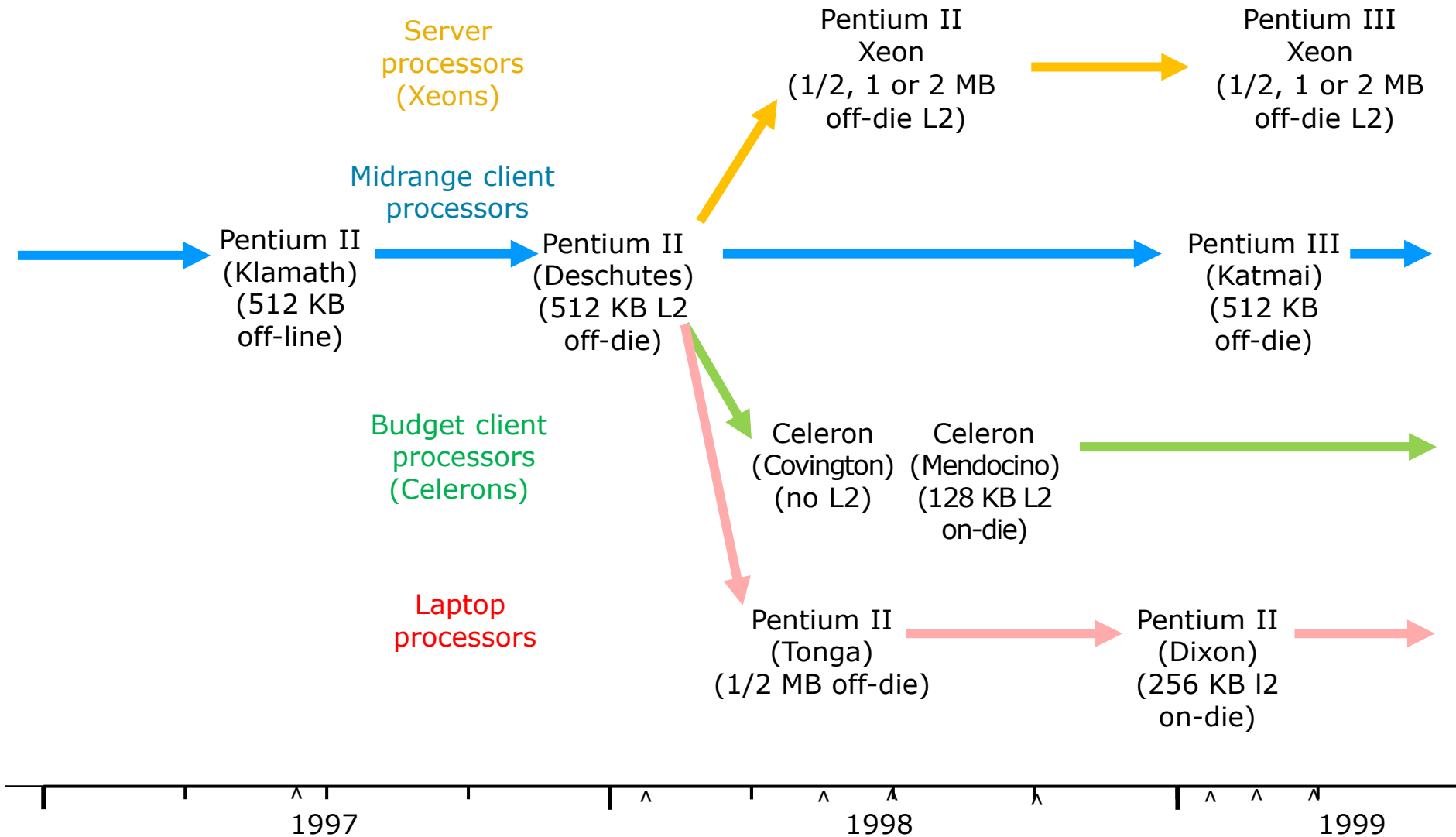
1. How did Intel become dominant in the processor market? (28)

IBM's, Apple's, and Motorola's 1991 alliance to challenge Intel's x86 supremacy -2

- At the beginning of the 1990s, Intel became overwhelmingly dominant in the microprocessor market, particularly with its x86 architecture.
- In a bold countermeasure, IBM, Apple, and Motorola formed an alliance to develop a competitive alternative.
- This collaboration resulted in the introduction of the 32-bit PowerPC line, which was a few years later expanded to a 64-bit line.
- While Apple integrated PowerPC processors into its Mac line and IBM launched PowerPC-based workstations and servers, the PowerPC architecture struggled to gain broader traction and was ultimately phased out from client devices by the mid-2000s.

1. How did Intel become dominant in the processor market? (29)

Intel's 1998 processor line split: a strategic move to better accommodate diverse market needs -1



1. How did Intel become dominant in the processor market? (30)

Intel's 1998 processor line split: a strategic move to better accommodate diverse market needs -2

- In 1998, Intel reshaped its processor strategy by introducing four distinct lines to better serve different market segments.
- This move was a response to the growing diversity in computing needs—from budget-conscious consumers to enterprise-level performance demands.
- Here's how Intel diversified its offerings beyond the Pentium II (Deschutes core):
 - server processors, known as Xeons
 - midrange client processors
 - budget client processors, known as Celerons and
 - mobile processors (essentially laptops in Intel's terminology).

1. How did Intel become dominant in the processor market? (31)

Recollection: The men who led Intel for 30 years



Photo: Intel's first CEOs: Noyce (1968-75), Moore (1975-87) and Grove (1987-1998) [128]

1. How did Intel become dominant in the processor market? (32)

Part of an interview with Andy Grove from 1999 [186]



Silicon Valley Historical Association

2. The race for dominance in the processor market

2. The race for dominance in the processor market (1)

2. The race for dominance in the processor market -1

- For over two decades, there has been fierce competition between Intel and AMD for dominance in the processor market.

- This competition focuses on three main battlefields, which are
 - the IC technology and
 - the performance leadership.

c) the processor efficiency.

Beyond IC technology and raw speed, it is the third crucial field of processor competition.

This metric indicates how much performance can be delivered per Watt consumed and is expressed in terms like GFLOPS/W.

Processor efficiency became the basic paradigm in designing processors since the middle of the 2000s.

- RISC architectures consume less power, therefore they are more power-efficient than their CISC-based x86 counterparts.

ARM processors are RISC-based, their power efficiency is the key factor behind ARM's dominance in the mobile market, where low power consumption and long battery life are essential.

In contrast, x86 processors are CISC-based, therefore they tend to consume more energy, making them less suitable for mobile environments.

2. The race for dominance in the processor market (1)

The race for dominance in the processor market -2

- In 2024, ARM-based Windows devices, primarily 2-in-1 laptops, or traditional laptops powered by highly efficient ARM-based Qualcomm processors entered the client market.
- This development posed stiff competition for both Intel and AMD, intensifying the processor battle on a third front, that focuses on the processor efficiency.
- Considering all three aspects mentioned above, the IC technology plays a decisive role because a firm relying on less advanced technology has a permanently lower chance to outperform its rival, which utilizes a more advanced process in processor performance or efficiency.
- In summary, there are three main battlefields in the processor war, which we will explore in this Section, followed by a discussion of Intel's recent market situation.

2. The race for dominance in the processor market

- 2.1 The race for technology leadership
- 2.2 The race for performance leadership
- 2.3 The race for efficiency leadership in laptop processors
- 2.4 What is the current market situation of Intel?

2.1 The race for technology leadership

2.1 The race for technology leadership

- To begin with, let's examine how Intel advanced its IC technology, commonly referred to as a process node, between 2000 and 2018, based on following Figure.
- The year 2000 saw the launch of the company's Pentium 4 processor line, while 2018 is the year when Intel introduced its long-awaited 10 nm technology, as will be discussed later.

2.1 The race for technology leadership (2)

Intel's advancements in IC technology up to the 10 nm process.

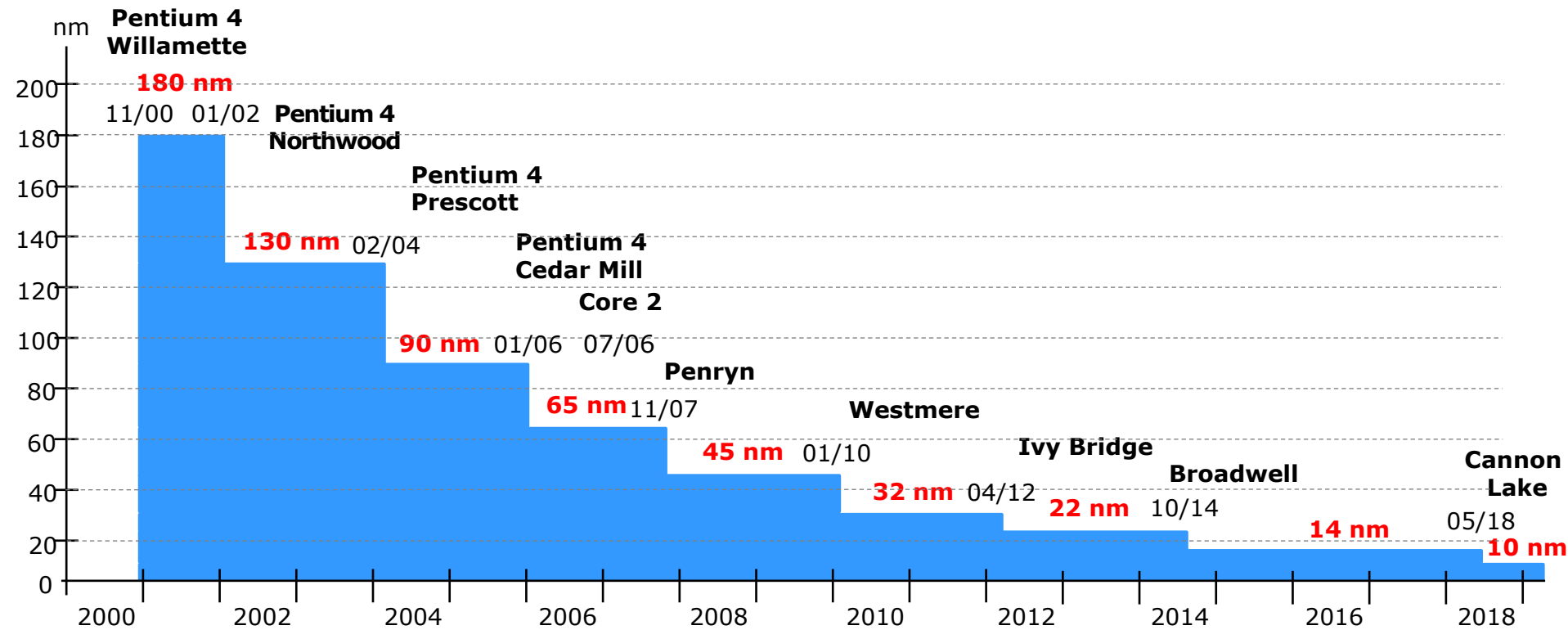


Figure: Intel's transitions to more advanced process technologies

- The Figure shows that Intel advanced its IC technology by shrinking its manufacturing process roughly **by 0.7 every two years**, until reaching the **14 nm technology**.
- Using a shrinking factor of **0.7** allows for **doubling** the number of transistors **on the same die area** every two years, in line with **Moore's (revised) law**.

2.1 The race for technology leadership (3)

Intel's leading role in the IC technology until the 14 nm feature size

Up to and including the 14 nm node, Intel consistently led the industry by introducing advanced IC technologies 2 to 4 years ahead of its competitors, as shown in the Figure below.

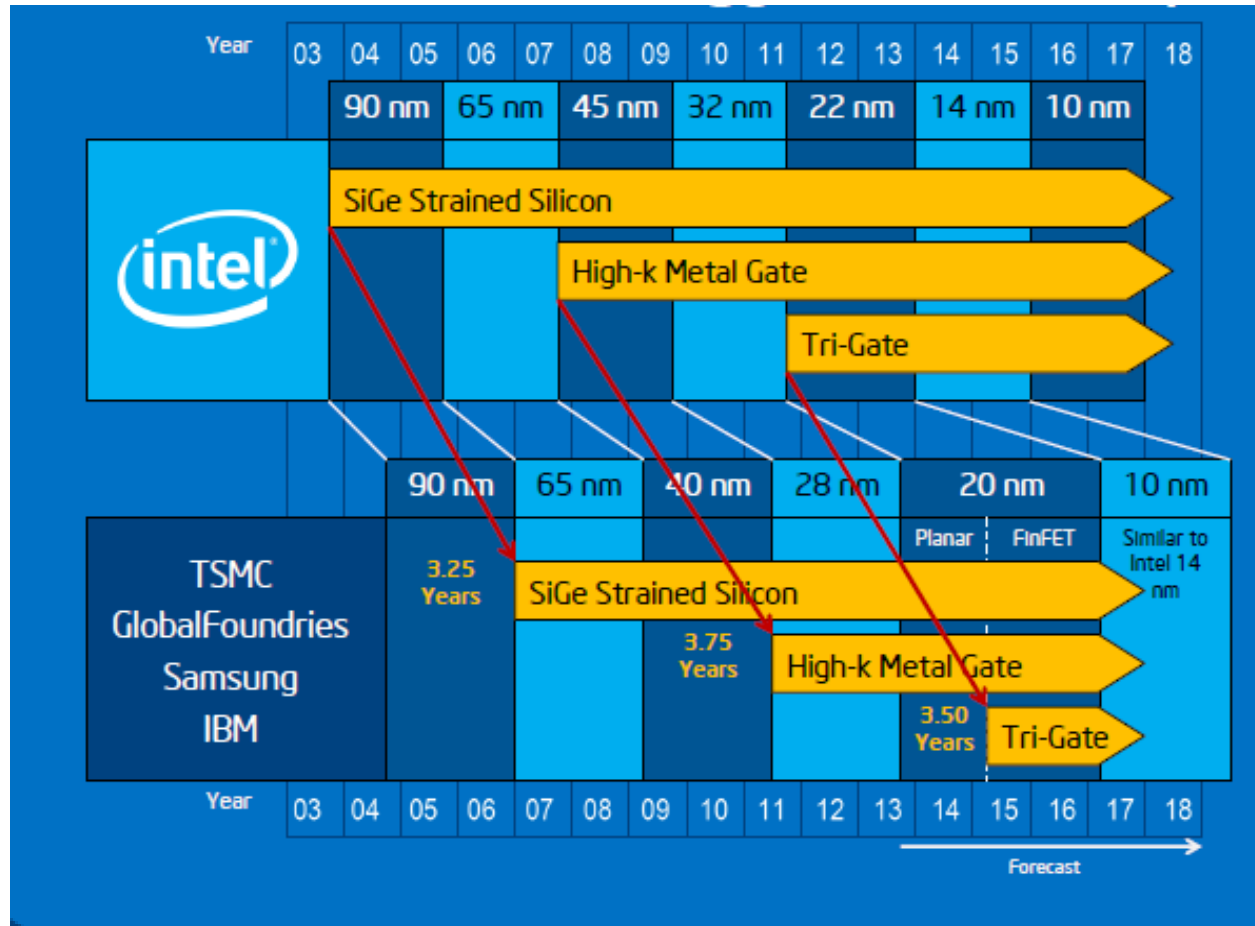


Figure: Intel's lead in introducing new transistor technologies up to the 14 nm node [1]

2.1 The race for technology leadership (4)

Intel's setback in transitioning to the 10 nm technology

- However, while advancing from 14 nm to 10 nm pilot production, Intel encountered unexpected challenges, taking nearly four years to achieve its target, as depicted in the Figure below.

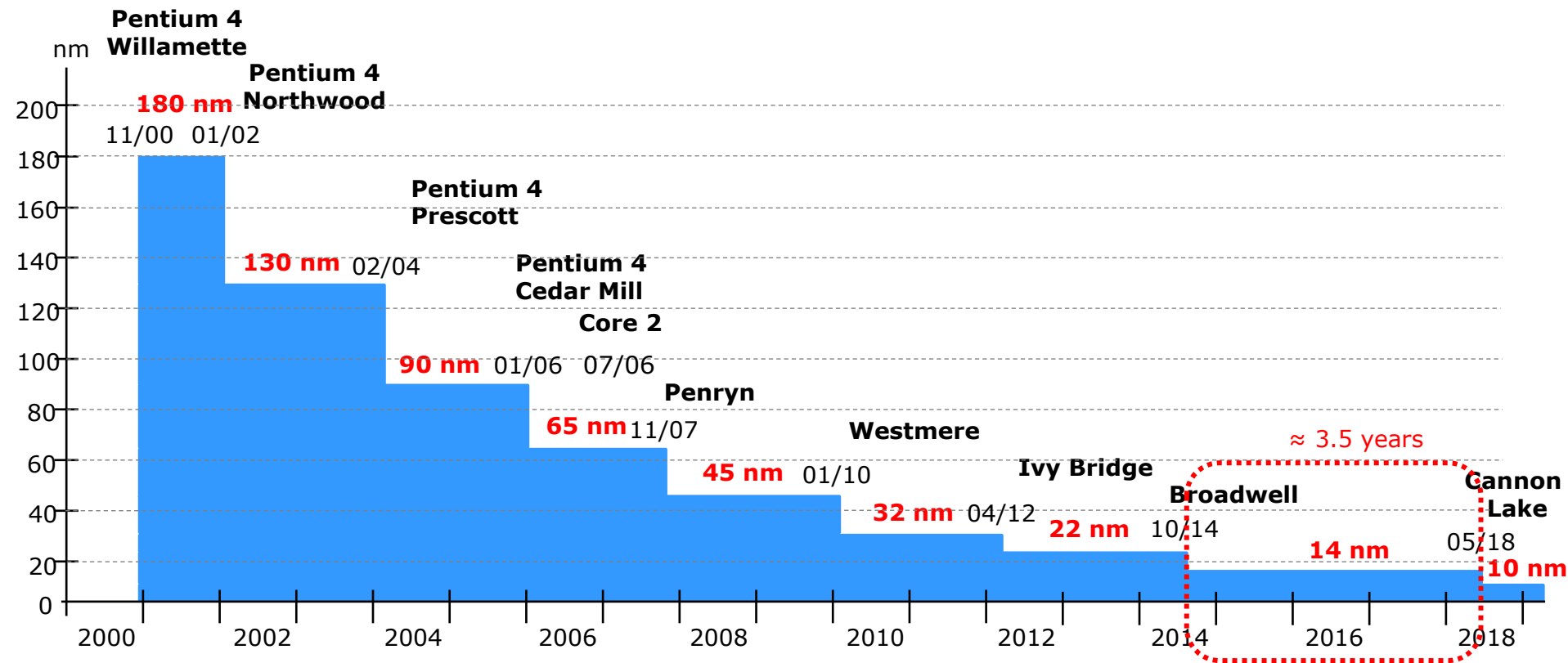


Figure: Intel's transitions to more advanced process technologies

- Meanwhile, competitors such as TSMC and Samsung caught up in IC development and eventually surpassed Intel, as illustrated by TSMC's progress in the following Figure.

2.1 The race for technology leadership (5)

TSMC's subsequent lead in process technology vs. Intel (stand: 2021) [2] -1
Below, we compare Intel's and TSMC's progress in IC manufacturing.

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
TSMC											
- Node	7	7/7+	5	5	3	3	2	2	1.5	1.5	
- Status	Risk	Full	Risk	Full	Risk	Full	Risk	Full	Risk	Full	
Intel											
- Node	14	14	10	10SF	10SF	7	7	5	5	3	3
- EN	13.8	13.8	7.1	7.1	7.1	4.1	4.1	2.4	2.4	1.3	1.3
- Status	Full	Full	Full	Full	Full	Ramp	Full	Ramp	Full	Ramp	Full

- Risk = risk starts, Ramp = production ramp, Full = full production, EN = TSMC equivalent node.

Table: Comparing Intel's and TSMC's nodes and their appearance [2]

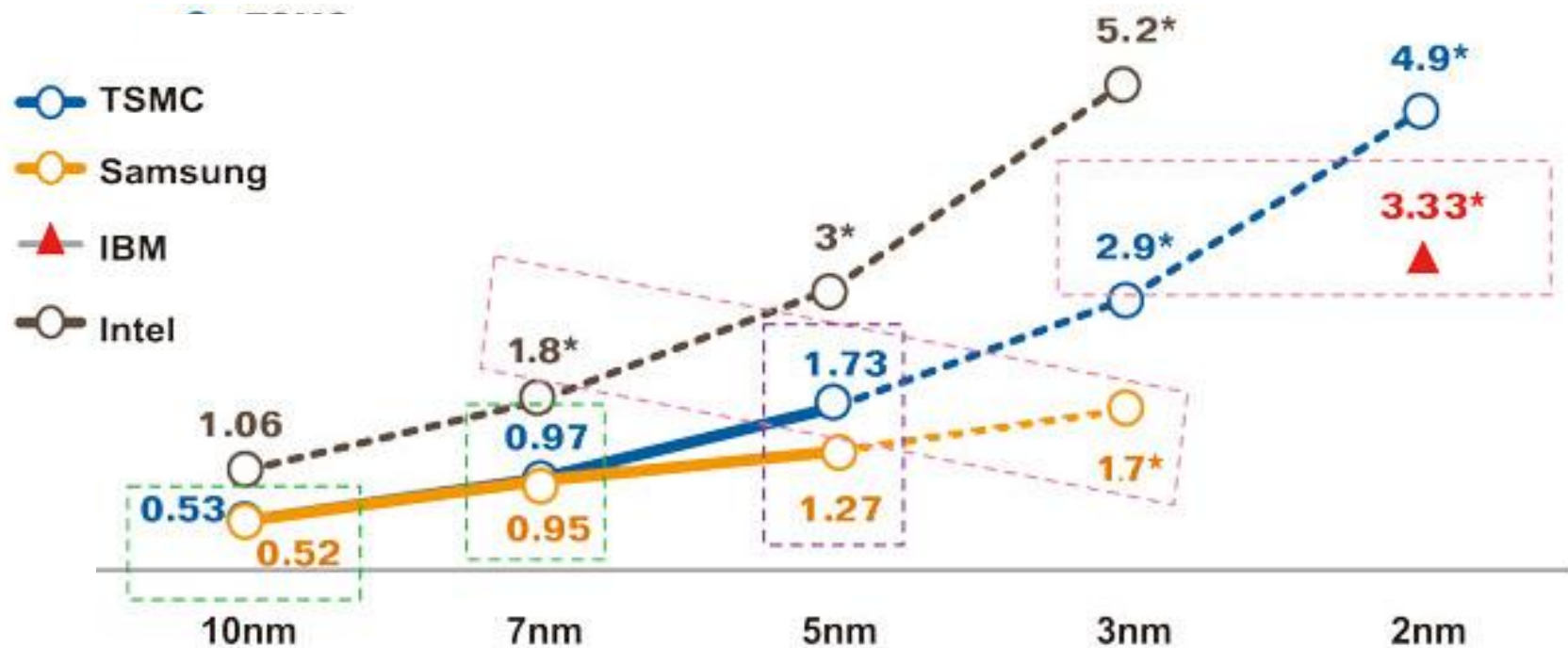
- The Table above compares the timelines of TSMC and Intel in introducing comparable generations of advanced IC technologies.
Comparable technologies are defined as those providing similar transistor densities.
- Based on this interpretation, the EN (equivalent) line indicates which TSMC node corresponds to each Intel node.

For instance, Intel's 10 nm node corresponds to TSMC's 7.1 nm node, etc.

- This alignment helps illustrate the shift in leadership in IC technology, with TSMC increasingly delivering equivalent or more advanced technologies one or two years ahead of Intel.

2.1 The race for technology leadership (6)

Transistor densities of the same nodes of major wafer foundries [3]
(Transistor densities are given in 100 million transistors/mm²)



- The graph indicates that recent IC processes deliver more than 100 million transistors/mm².
 - Note also that Intel's recent IC processes demonstrate transistor densities comparable to those of TSMC's or Samsung's next-generation IC processes.
 - For instance, Intel's 10 nm technology achieves a transistor density of 106 million transistors/mm², which is on par with the densities of TSMC's and Samsung's 7 nm processes (97 and 95 million transistors/mm², respectively).
- Likewise, Intel's 7 nm process offers a transistor density that is roughly equivalent to TSMC's 5 nm or Samsung's 3 nm processes.

2.1 The race for technology leadership (7)

Intel's new process naming scheme from 07/2021 [4]

- This situation is **disadvantageous for Intel** because its process node naming convention **exhibits larger nm values than TSMC or Samsung for similar transistor densities**.

For example, Intel's 10 nm process achieves a transistor density comparable to TSMC's 7 nm process, which may create a misleading perception that Intel's technology is less advanced.

- To address the disadvantage of its address convention, **Intel** introduced a **new naming scheme** for its IC manufacturing processes (commonly referred to as nodes).
- Under this new scheme **each node name begins** with "**Intel**" **followed by a number** that **corresponds to TSMC's equivalent process designation, ensuring the same density**.
- Importantly, these names **omit the nm suffix** to avoid misleading comparisons.

For instance, Intel's former 10 nm process is now referred to as "Intel 7", reflecting its parity with TSMC's 7 nm node in terms of transistor density.

2.1 The race for technology leadership (8)

Intel's new process naming scheme from 07/2021 [4]

Previously
referred to

10 nm
Enhanced Superfin

7 nm

10nm
SuperFin

**Intel
7**

**Intel
4**

**Intel
3**

**Intel
20A**

Previously referred to
as Enhanced SuperFin

- In high-volume production

- 10-15% perf/watt gain
- FinFET transistor optimizations
- Now in volume production

Previously referred
to as 7nm

- 20% perf/watt gain
- Full use of EUV lithography
- Meteor Lake for client tape in Q2 2021
- Granite Rapids compute tile for data center

Power and area
improvements

- 18% perf/watt gain
- Denser HP library
- Increased intrinsic drive current
- Reduced via resistance
- Increased EUV use
- Manufacturing products 2H 2023

The angstrom era
of semiconductors

- Breakthrough innovations in 1H 2024
- RibbonFET – new transistor architecture
- PowerVia – interconnect innovation

2.1 The race for technology leadership (9)

Intel's ambitious process technology roadmap as of 12/2019 [5]

To catch up in IC technology, Intel unveiled an **ambitious IC manufacturing plan** in 12/2019, aiming to transition through **five nodes within 10 years**, as shown below.

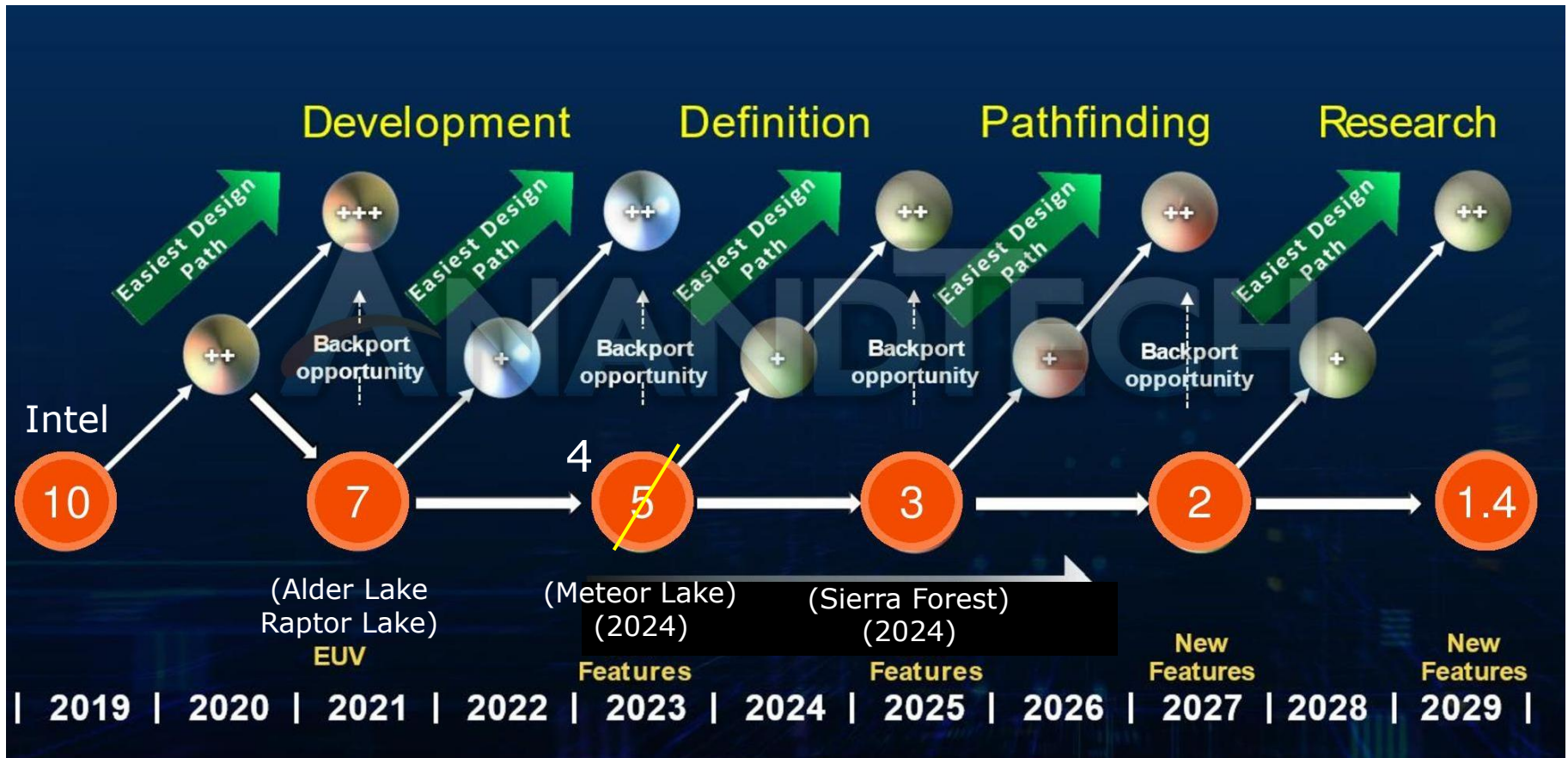


Figure: Intel's process technology roadmap from 12/2019 [5]

EUV: Extreme ultraviolet lithography

2.1 The race for technology leadership (10)

Intel's accelerated, over-ambitious process technology roadmap as of 09/2023 [6]

In [9/2023](#), Intel made a surprising announcement, unveiling an **updated process roadmap** that revealed **significantly earlier completion targets** for its upcoming Intel 3 and Intel 20A process technologies by **2 to 3 years ahead of previous projections**, as shown below.

5 NODES IN 4 YEARS



Figure: Intel's updated process technology roadmap from 09/2023 [6]

2.1 The race for technology leadership (11)

Delays in implementing Intel's over-ambitious process roadmap as of 12/2024 [6]



As illustrated in the Figure, **Intel is facing significant setbacks in advancing its process technology.**

Compared to its accelerated roadmap announced in 09/2023, the company is now **trailing by approximately one year.**

2.1 The race for technology leadership (12)

Remarks to Intel's implementation of the A20 and A18 processes

- An indication of Intel's ongoing difficulties in advancing its process technology is the reliance on TSMC's advanced nodes for its latest products.

Both the Lunar Lake laptop processors, launched in 09/2024, and the Arrow Lake desktop processors, released in 10/2024, were fabricated using TSMC's N3B, N5P, and N6 technologies – rather than Intel's originally planned 18A and 20A processes.

- Further underscoring Intel's difficulties in mastering advanced technologies, industry sources report that Intel's 18A process achieved a very low yield rate of less than 10 % in 12/2024 [137].

2.1 The race for technology leadership (13)

A contrast: TSMC's process technology roadmap as of Q2/2022

After showcasing Intel's process roadmap from 09/2023, it becomes challenging to display TSMC's technology roadmap as of Q2/2022 as well (see below).

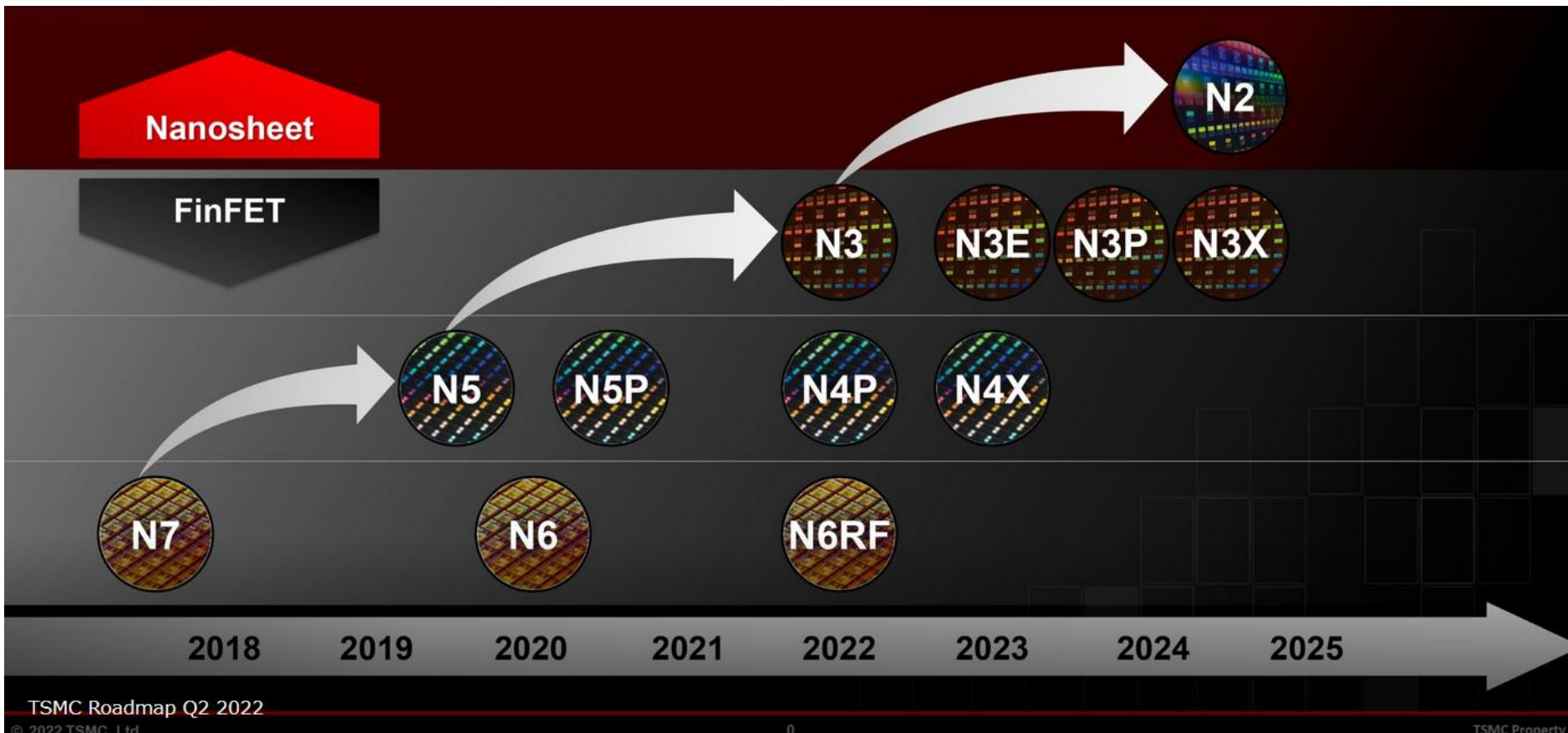


Figure: TSMC's process technology roadmap from Q2/2022 [7]

N: Node E: Enhanced P: Performance X: Extreme performance

2.1 The race for technology leadership (14)

Interpretation of the tags used by TSMC in its process technology roadmap

- TSMC distinguishes its more advanced processes by using tags.
- The process tags are interpreted as follows:
 - E: Enhanced
 - P: Performance
 - X: Extreme performance
- Upcoming processes provide higher transistor density and clock frequency, as indicated below.

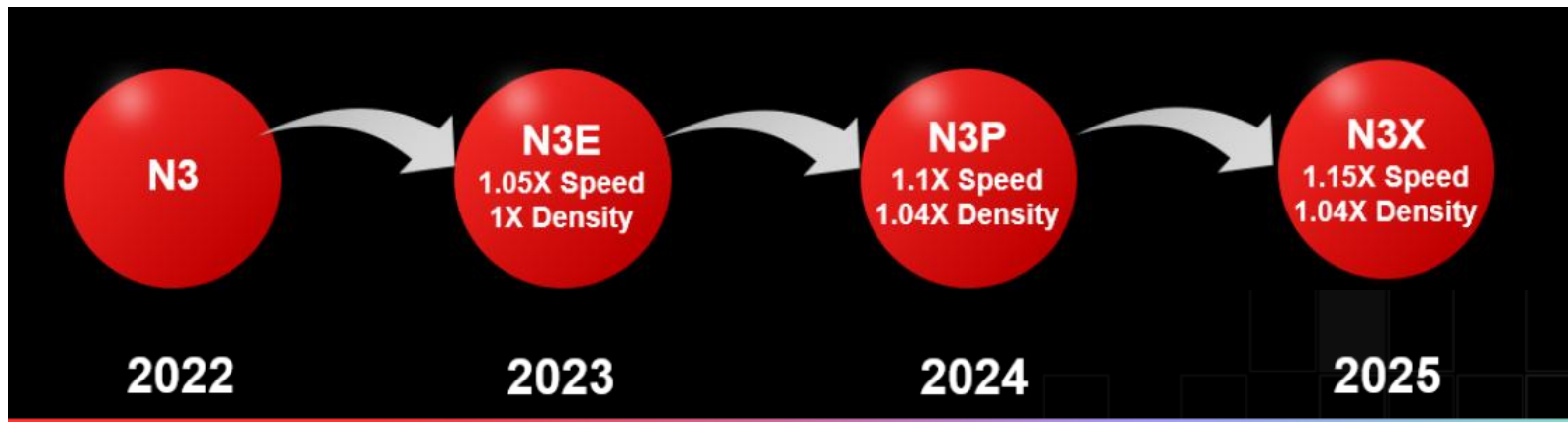
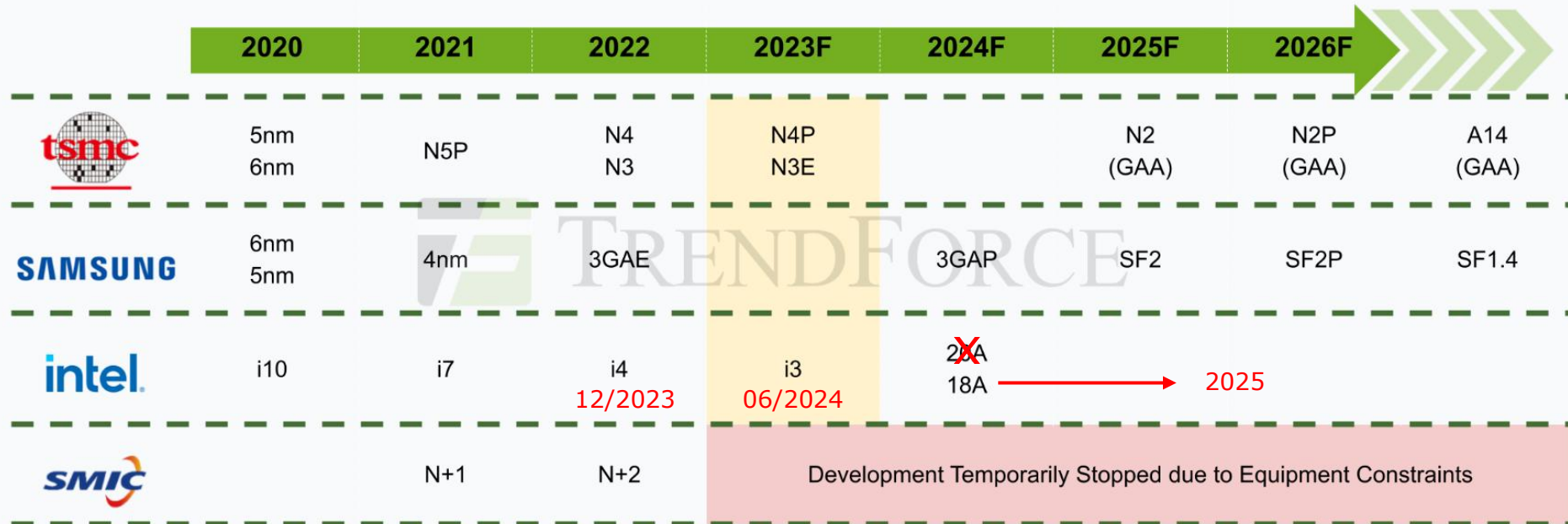


Figure: Speed/density advancements of TSMC's subsequent N3 processes [8]

2.1 The race for technology leadership (15)

Comparing the roadmaps of major foundries, as of 09/2023 [138]



Source: TrendForce, Sep., 2023

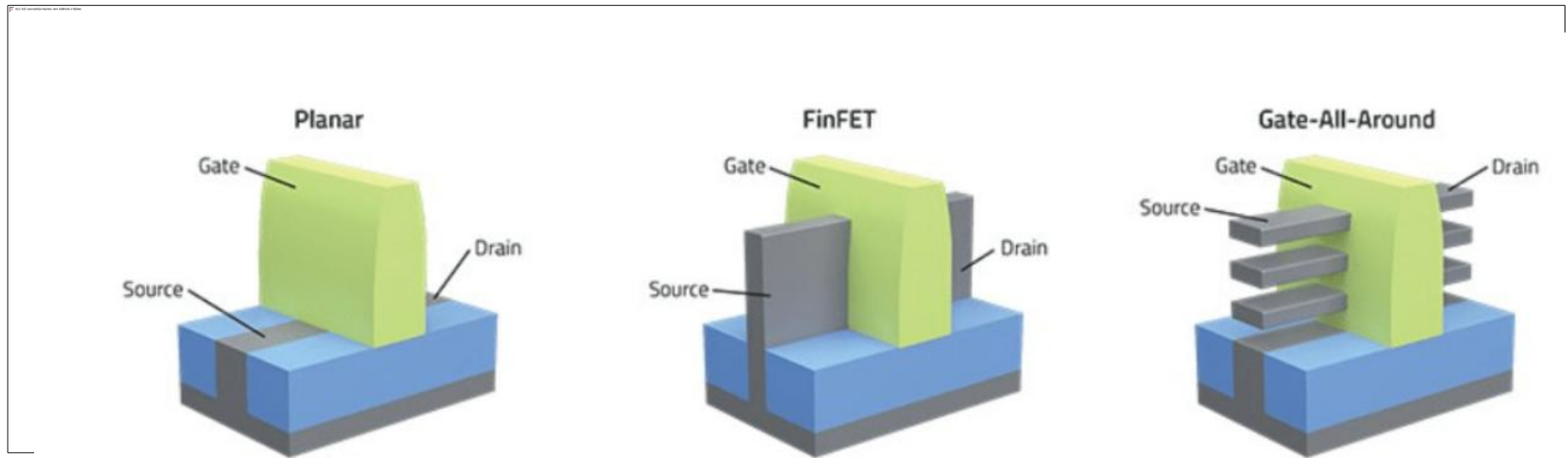
- Available publications [208, [209] indicate that by the end of 2025, both TSMC and Samsung began producing 2 nm dies.
- This suggests that at that time, Intel, with its 1.8A process, could catch up with its competitors TSMC and Samsung.

GAA: Gate-All-Around (kaput körülölő) transistor technology (see the following Figure).

SMIC: Semiconductor Manufacturing International Corporation
(Largest semiconductor foundry in mainland China, based in Shanghai.)

2.1 The race for technology leadership (16)

Remark: Emergence of the Gate-All-Around (GAA) transistor technology [139]



1959

Introduced by Intel in 2012 along with the 22 nm Ivy Bridge

All major foundries are developing different kinds of the GAA technology, such as Samsung's GAE, Intel's, 18A or TSMC's N2 technology.

The **planar** topology of FETs allowed the design and manufacture of integrated circuits.

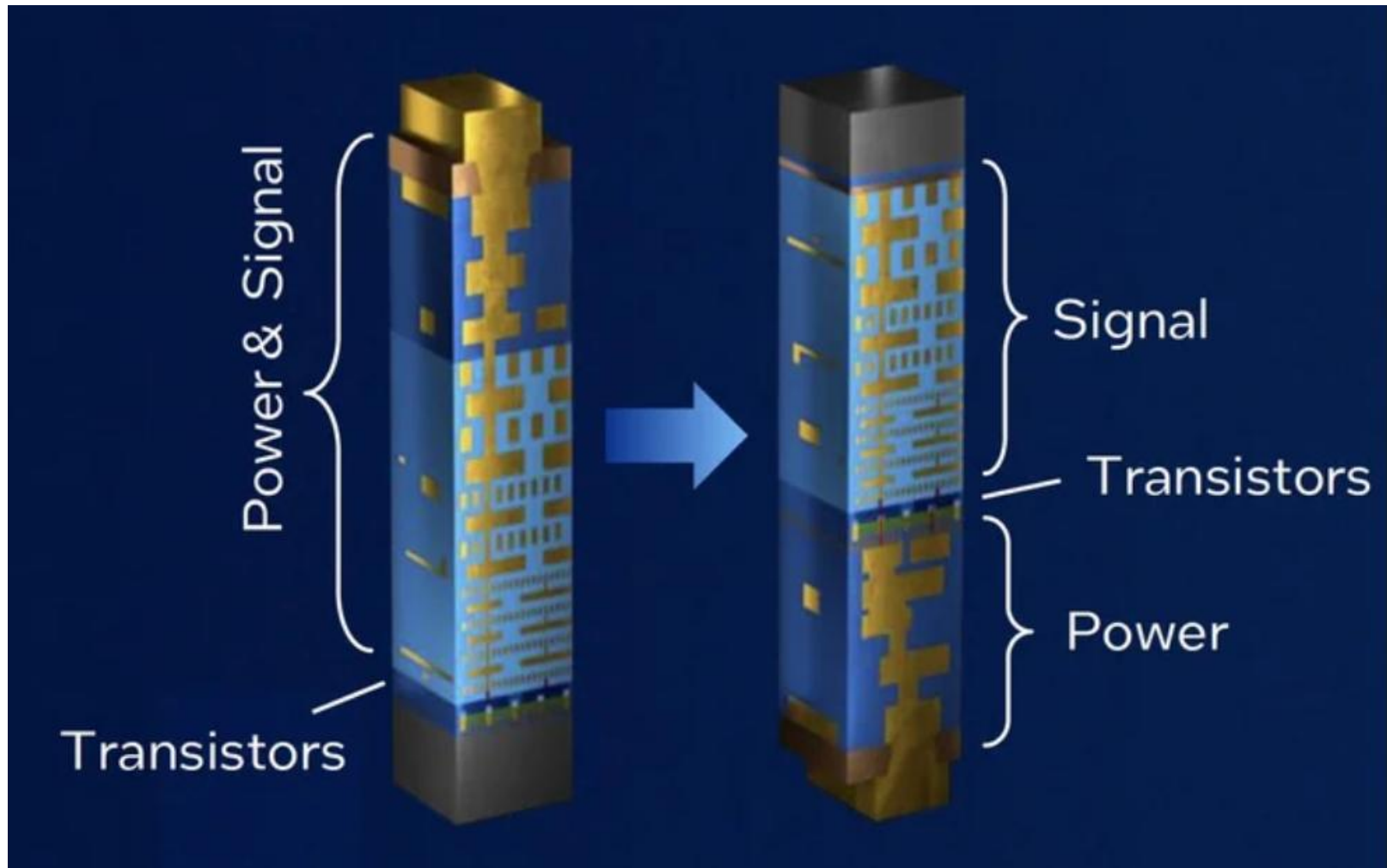
FET: Field Effect Transistor (térvezérlésű tranzisztor) replaced bipolar junction transistor.

FinFET: Fin Field-Effect Transistor

In the **GAA transistor technology**, the **gate completely surrounds the channel**, providing better control over the current flow and reducing leakage current.

2.1 The race for technology leadership (17)

A further innovation of Intel's 18A technology: back-side power delivery [198]



- Traditionally, signal and power interconnects are mixed.
- With **back-side power delivery**, they are **separated**, as seen above.

2.2 The race for performance leadership

2.2 The race for performance leadership (1)

2.2 The race for performance leadership -1

Who has the fastest processors, Intel or AMD?



Image source: [12]

b) The race for performance leadership -2

- Intel and AMD have long been fierce rivals in the processor market.
- From approximately 2003 to 2005, AMD surpassed Intel in processor performance thanks to the innovations introduced.

These included extending the x86 ISA to 64-bit and integrating the memory controller directly onto the processor die, as detailed in the Chapter on Zen family.

- The following Figure presents related performance data in terms of Cinebench R15 scores.

Remark

AMD's first 64-bit processor was known as *Opteron*, it introduced direct-connected memory as well.

It was officially released in 04/2003.

Remarks on the Cinebench benchmark program

- Cinebench is a widely used cross-platform benchmarking suite designed to evaluate processor performance by rendering complex 3D scenes.

It utilizes both CPU and GPU capabilities to simulate real-world workloads.

- Developed by MAXON, a company known for its expertise in animation and visual effects software.

The designation Cinebench comes from Cinema and Benchmark.

- Cinebench has evolved through several major versions—such as R15, R20, and R23—each released in different years with updated testing methodologies.

Note that scores from different Cinebench versions (e.g., R15 vs. R23) are not directly comparable, as each version introduces changes in rendering complexity and system demands.

- Both Intel and AMD frequently use Cinebench results to showcase and compare the performance of their processors.

2.2 The race for performance leadership (4)

AMD's lead in the processor performance race around 2003-2006

The Figure below displays **Cinebench R15 scores** for the period when AMD's K8-based processors outperformed Intel's Pentium 4 (P4) counterparts.

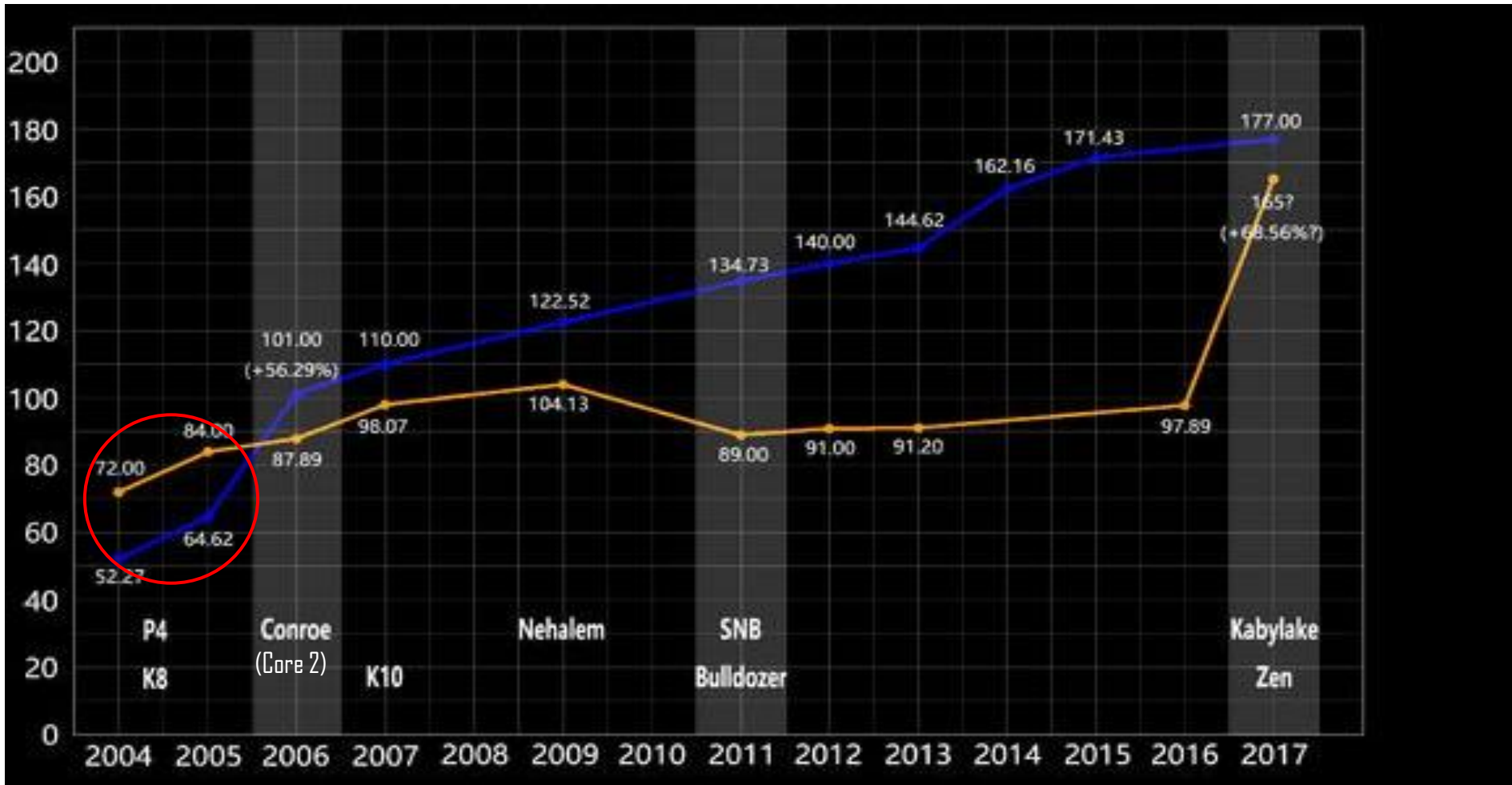


Figure: Cinebench R15 Single core performance results @ 4.0 GHz [13]

2.2 The race for performance leadership (5)

Intel's rise to the processor performance leadership -1

With Intel's **Core 2** processors, a new era began in which the company **dominated the CPU performance battle for nearly 15 years**, as illustrated in the Figure below.

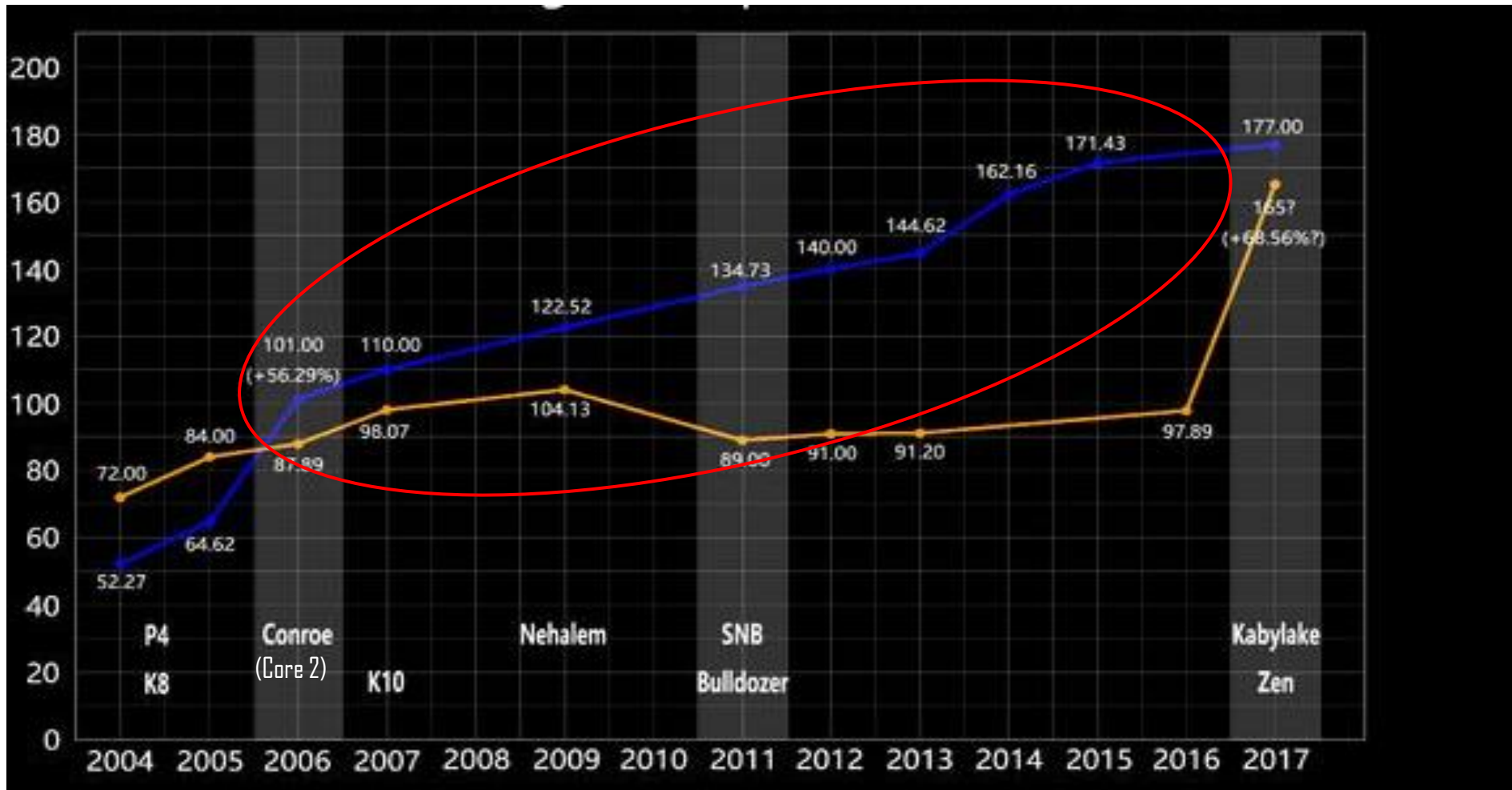


Figure: Cinebench R15 Single core performance results @ 4.0 GHz [13]

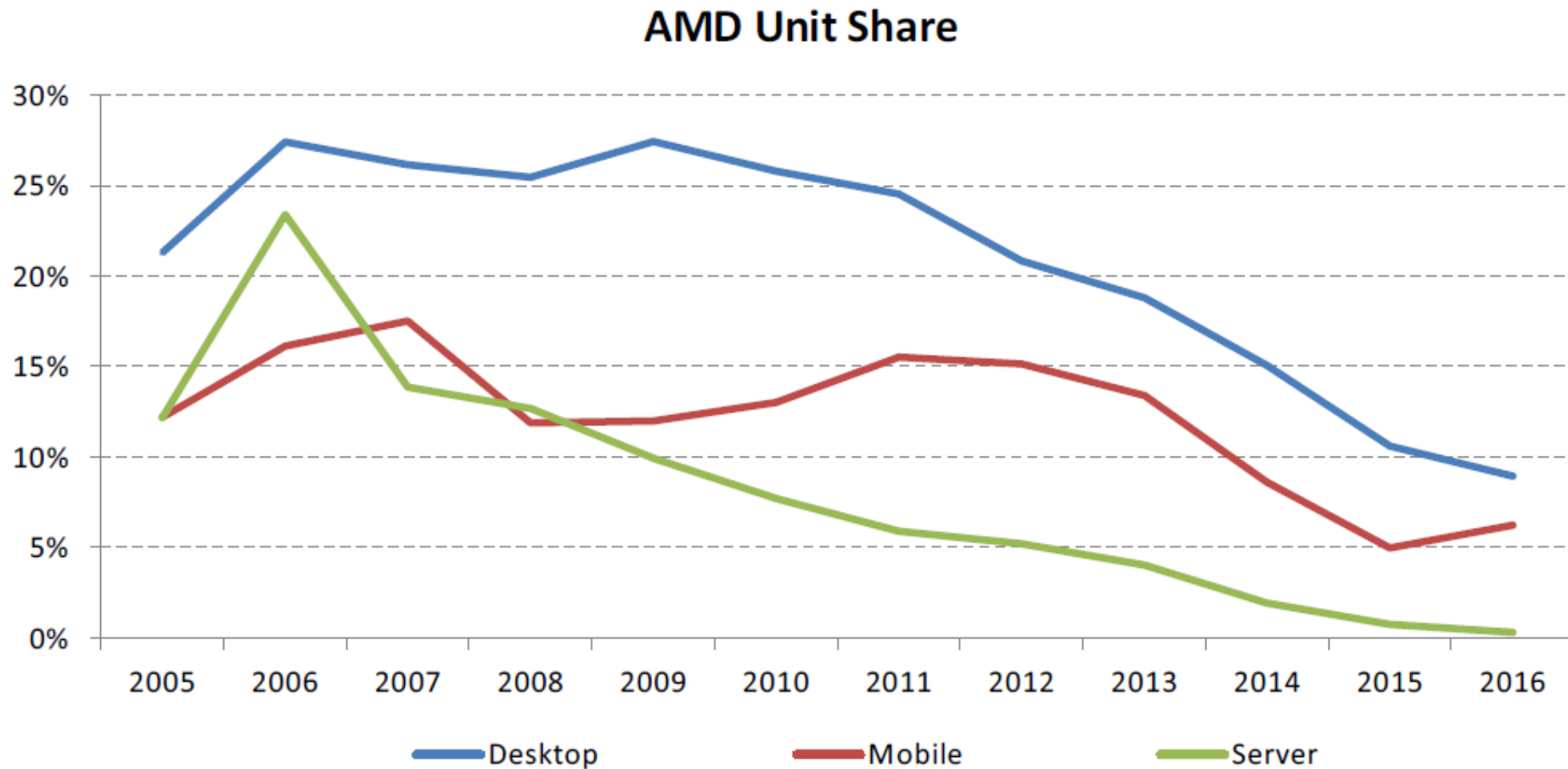
Intel's rise to the processor performance leadership -2

At this time Intel had

- better process technology than AMD (65 nm vs. 90 nm) and
- wider cores, as Core 2-based processors were 4-wide - while AMD's K8-based processors only 3-wide.

2.2 The race for performance leadership (7)

Intel vs. AMD: How the widening performance gap led to AMD's shrinking global market share [14]



Source: Mercury, Morgan Stanley Research

2.2 The race for performance leadership (8)

A game changer in the processor race: introduction of AMD's Zen family

In 2017, AMD launched its Zen processor family, with cores that deliver 52 % higher-performance than the previous Bulldozer family, which gave rise to a game change [13]

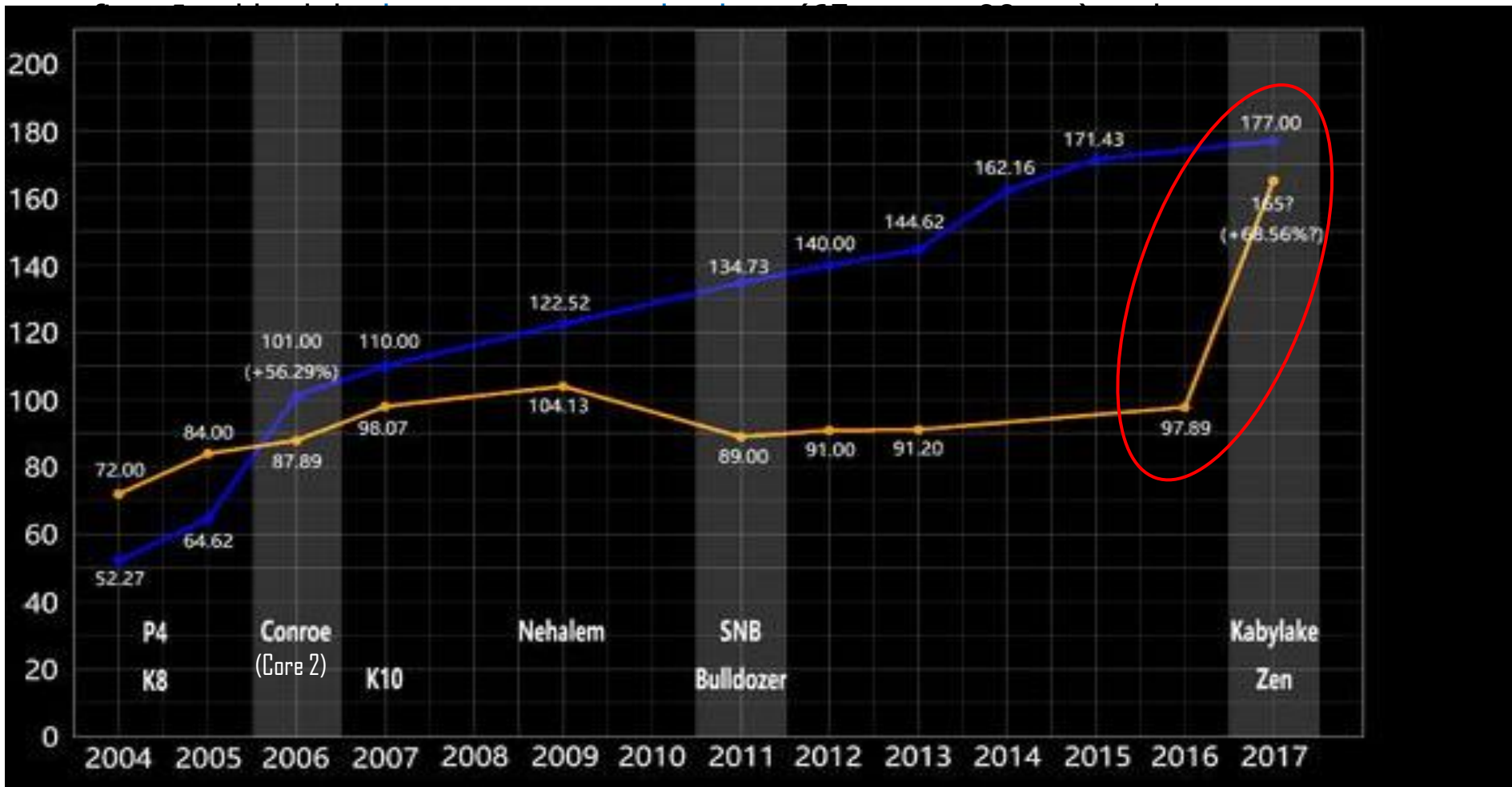


Figure: Cinebench R15 Single core performance results @ 4.0 GHz [13]

2.2 The race for performance leadership (9)

AMD's regained lead in the performance race with their Zen 3 processors

In 11/2020, AMD stated that its Zen 3-based gaming DT processors had **surpassed** Intel's 10th gen. DT chips in many aspects, marking a breakthrough after nearly 15 years of trailing behind, as shown below [15].

Metric	Market Leader
1080p Gaming	AMD 
Core Architecture IPC	AMD 
Power Efficiency	AMD 
Single-Thread Performance	AMD 
Multi-Thread Performance	AMD 
Performance per Dollar	AMD 
Backwards Compatibility	AMD 

Table: Comparing various performance metrics of the Zen 3-based Ryzen 9 5900X DT processor vs. Intel's 10th gen. Core i9-10900K [15]

- This announcement served as a **powerful marketing message** for AMD, especially **in contrast to Intel's earlier claim** (04/2020) that **its new, 10-core 10th gen. DT processor was the best gaming CPU available on the market.** IPC: Instructions Per Cycle

2.2 The race for performance leadership (10)

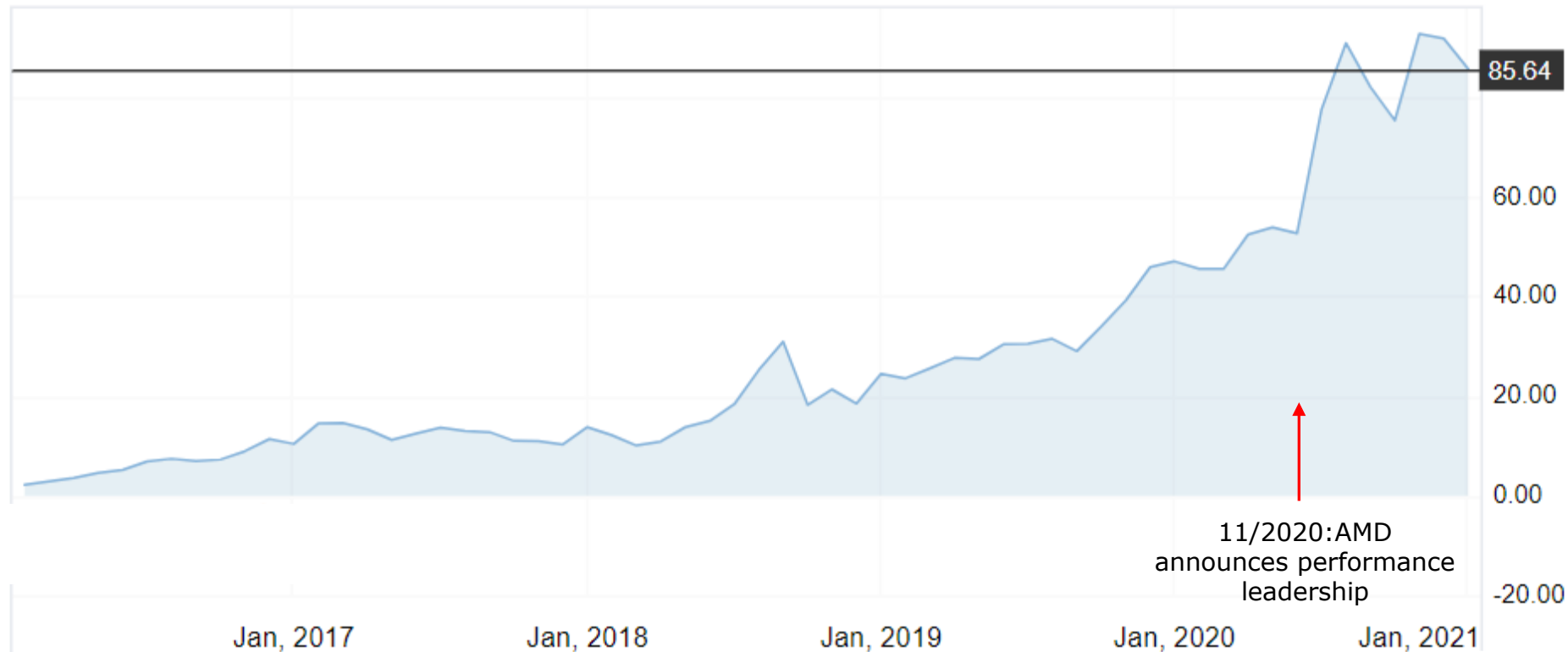
A quantitative illustration of AMD's performance claim:
Single-thread Cinebench R20 scores of the referenced processors [15]

Cinebench R20 Rendering Benchmark - Single Thread Scores				
	uArch	Process	1C Turbo	Score
AMD Ryzen 9 5900X	Zen3	TSMC 7nm	4.8 GHz	631pts
Intel Core i9-10900K	Comet Lake	Intel 14+++	5.0 GHz	538pts
AMD Ryzen 9 3950X	Zen2	TSMC 7nm	4.7 GHz	536pts

- Based on the data above, AMD's Zen 3-based Ryzen 9 5900X significantly outperforms Intel's flagship Core i9-10900K in terms of Cinebench R20 single-thread performance.
- There are **two main reasons** for this
 - a) Superior process technology
At this time AMD had a better process technology using TSMC's 7 nm process compared to Intel's aging 14+++ technology and
 - b) More efficient microarchitecture
- AMD's Zen 3 microarchitecture brought notable improvements in IPC.
Albeit Intel's Core i9-10900K has 5-wide cores, AMD's Zen 3 cores are still 4-wide, AMD's microarchitectural refinements allowed to surpass Intel in processor performance.

2.2 The race for performance leadership (11)

The remarkable rise of AMD's stock price between 2017 and 2021



According to the data above, [AMD's stock price skyrocketed](#) from just \$2 in 2016 to over \$80 by 2020.

This dramatic rise was fueled by the company [reclaiming the processor performance crown](#) with its Zen 3 architecture, which also led to a significant increase in market share.

2.2 The race for performance leadership (12)

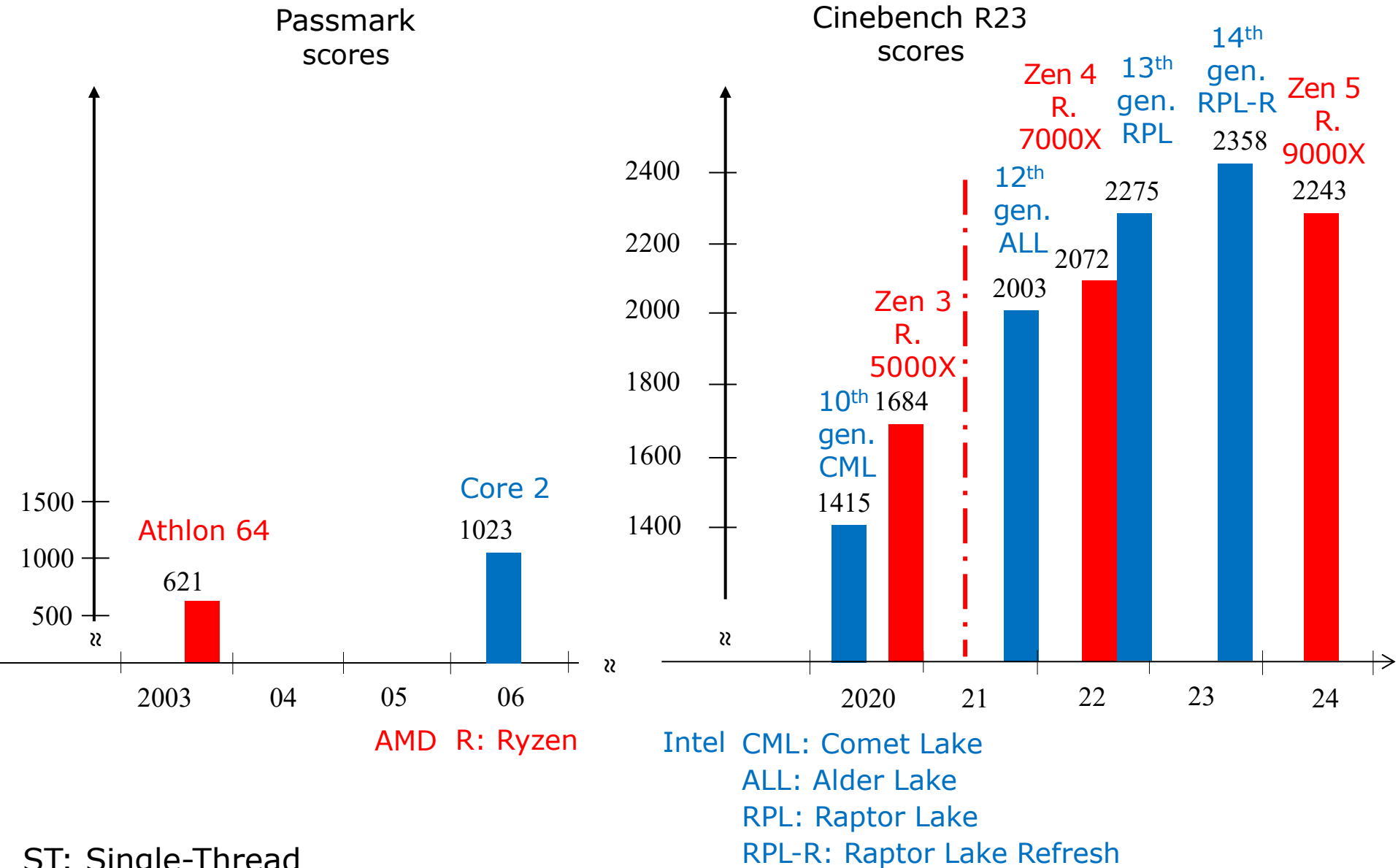
What's happened next in the processor performance race between Intel and AMD?

- Next, a fierce competition has emerged between Intel and AMD, with each company alternately claiming the performance leadership,

as the following Figure demonstrates for gaming DTs.

2.2 The race for performance leadership (13)

The gaming DT processor performance race: ST performance - Intel vs. AMD



Alternating leadership in the gaming DT performance race: Intel vs. AMD

The Figure above illustrates how Intel and AMD have **alternated** in leading the gaming desktop performance race.

The recent performance ranking of server and client processors

It is challenging to gain a clear overview of the recent competitive landscape in both
the client and server processor performance.

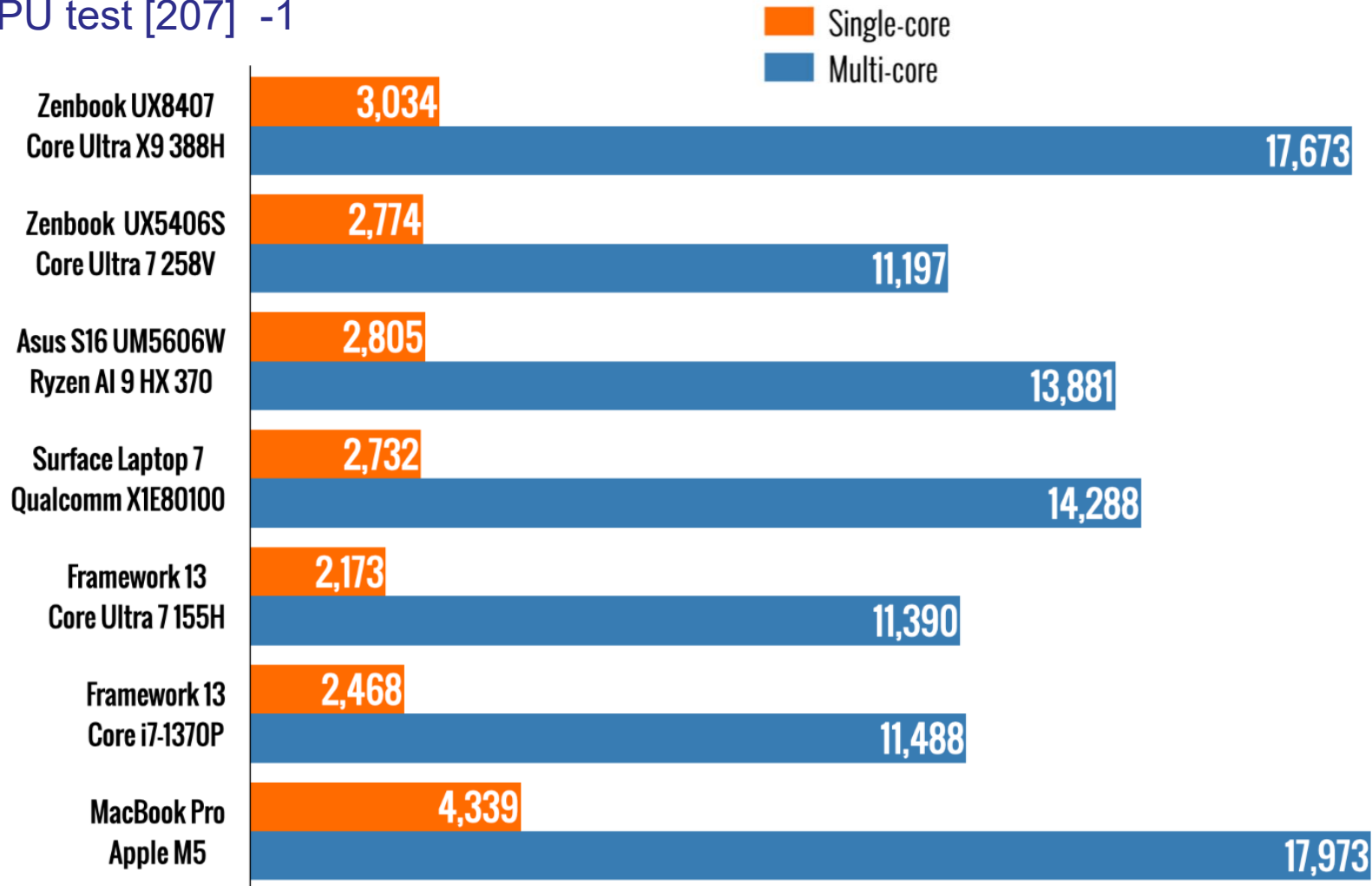
The following two Subsections address this point.

a) Recent developments in laptop processor performance leadership

- In 01/2026, Intel released its Core Ultra Series 3 processors, known as Panther Lake.
- The CPUs of these processors are manufactured using Intel's cutting-edge 1.8A technology.
- This processor Series demonstrates extraordinary performance, surpassing all SC and MC performance figures of competing recent laptop processors, with the exception of Apple's M5, as shown in the following Figure.

2.2 The race for performance leadership (17)

SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [207] -1



Remark

Geekbench 6 is cross-platform benchmark that evaluates how a CPU handles everyday tasks such as email, photo operations, music playback, augmented reality, and machine-learning-style computations.

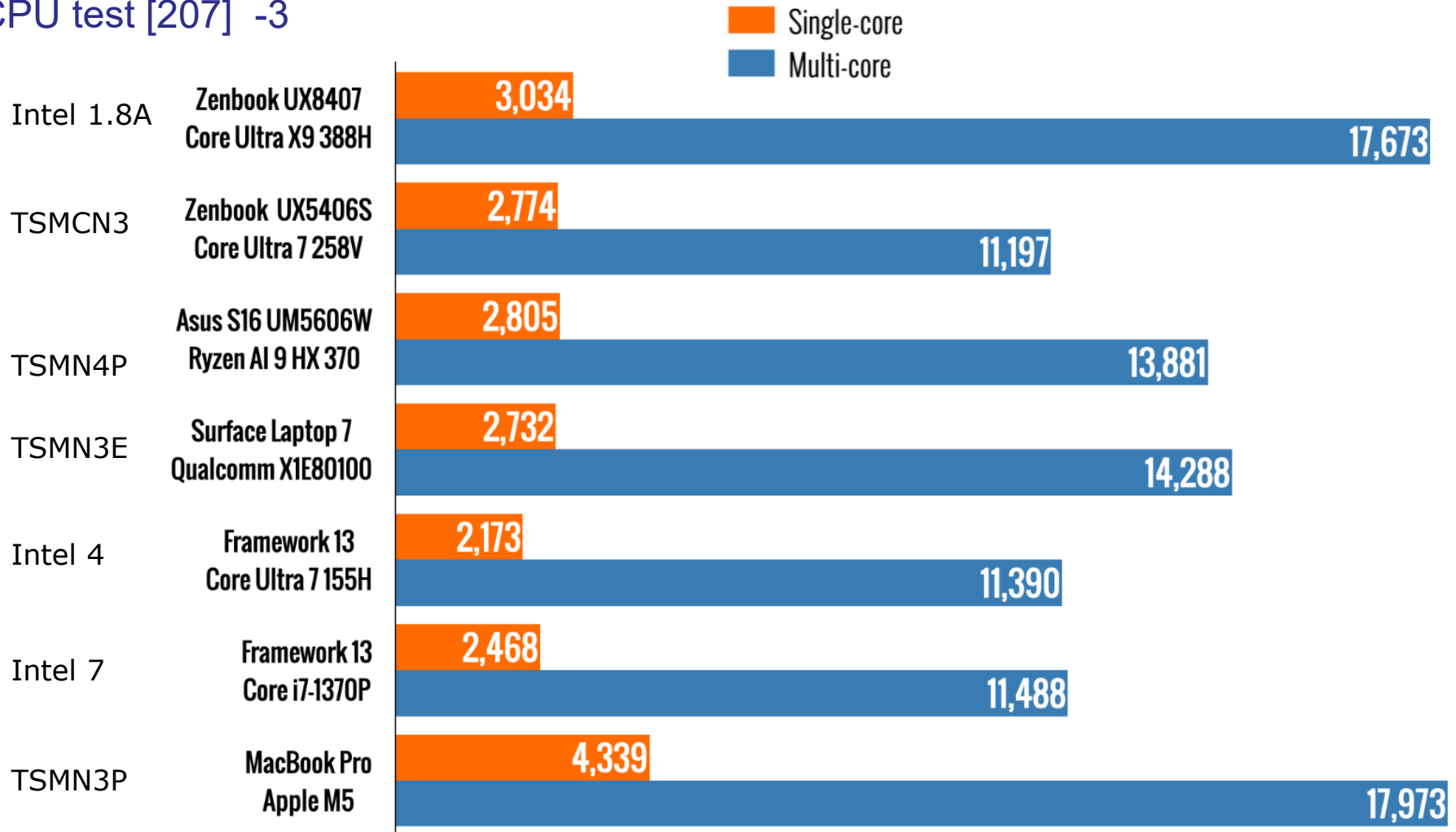
SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [207] -2

While assessing the above results **caution is advised**.

Intel's Panther Lake was fabricated using the latest advanced manufacturing process (Intel 1.8A), in contrast, the CPUs of the indicated competitors utilize less advanced technologies, primarily 3 nm or 4 nm processes, as shown below.

2.2 The race for performance leadership (19)

SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [207] -3



Remark

Geekbench 6 is a cross-platform benchmark that evaluates how a CPU handles everyday tasks such as email, photo operations, music playback, augmented reality, and machine-learning-style computations.

2.2 The race for performance leadership (20)

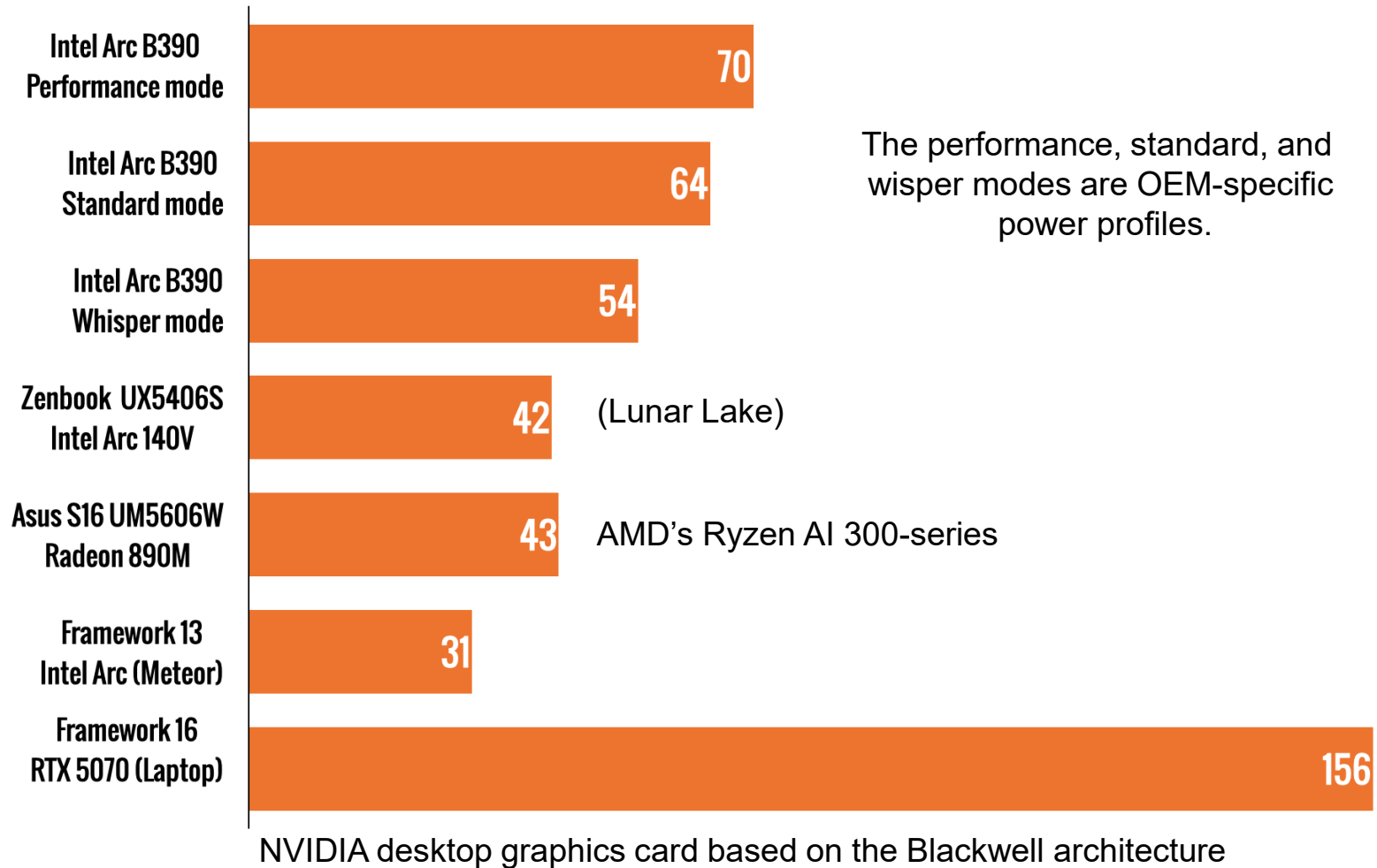
SC and MC performance of the Panther Lake processors using the Geekbench 6 CPU test [207] -4

A more advanced manufacturing process, however, offers about 10 -30 % higher performance and lower power consumption than earlier processes [209].

Consequently, new processors from Qualcomm or AMD, which will use more advanced or similar manufacturing processes may outperform Intel's Panther Lake processors in terms of performance or power consumption.

2.2 The race for performance leadership (21)

Graphics performance comparison while running the Shadow of the tomb raider game (Results in FPS, higher is better) [207] -1



It is a [single-player action-adventure game](#), introduced in 2018.

In the story, Lara Croft races to stop a Maya apocalypse while exploring jungles, tombs, and underwater environments.

2.2 The race for performance leadership (22)

Graphics performance comparison while running the Shadow of the tomb raider game (Results in FPS, higher is better) [207] -2

- Intel's Core Ultra Series 3 processors once again deliver the highest performance among integrated graphics.
- Clearly, the RTX 570 graphics card outperforms integrated graphics largely.
- Regarding graphics performance, we emphasize that Panther Lake's graphics utilize more advanced IC technology compared to the competition, which in turn raises concerns, as previously discussed.

Update on the performance competition in the laptop segment

- In 02/2026, several benchmark results were released for **Qualcomm's Snapdragon X2 family**, announced in 09/2025 but not yet launched.
It is expected to be released in H1 2026.
- This high-performance ARM-based laptop line utilizes a **big-little architecture** with **up to 12 performance and 6 efficiency cores**.
- It is fabricated using **TSMC's 3 nm** technology.
- While the published benchmark results are impressive, there is currently **no reliable information available regarding power consumption or battery life**.
- Below, we present a comparison of benchmark results, including the three highest-performing laptop processors from Intel, Apple, and Qualcomm.

2.2 The race for performance leadership (24)

Comparing benchmark results of Intel's, Apple's, and Qualcomm's highest-performing laptop processors (as of 02/2026) [211] -1

	Snapdragon X2 Elite	Apple M5	Intel Panther Lake (CU7)
Cinebench 2024 (Single)	146	200	~131*
Cinebench 2024 (Multi)	1,432	1,153	~1,210*
Handbrake (Video Transcode)	3:29	3:31	4:32
Blender 5.01 (Render Time)	3:31	5:33	5:10
Geekbench 6 (Multi)	23,449 (Extreme)	17,926	16,927
*Estimated			

2.2 The race for performance leadership (25)

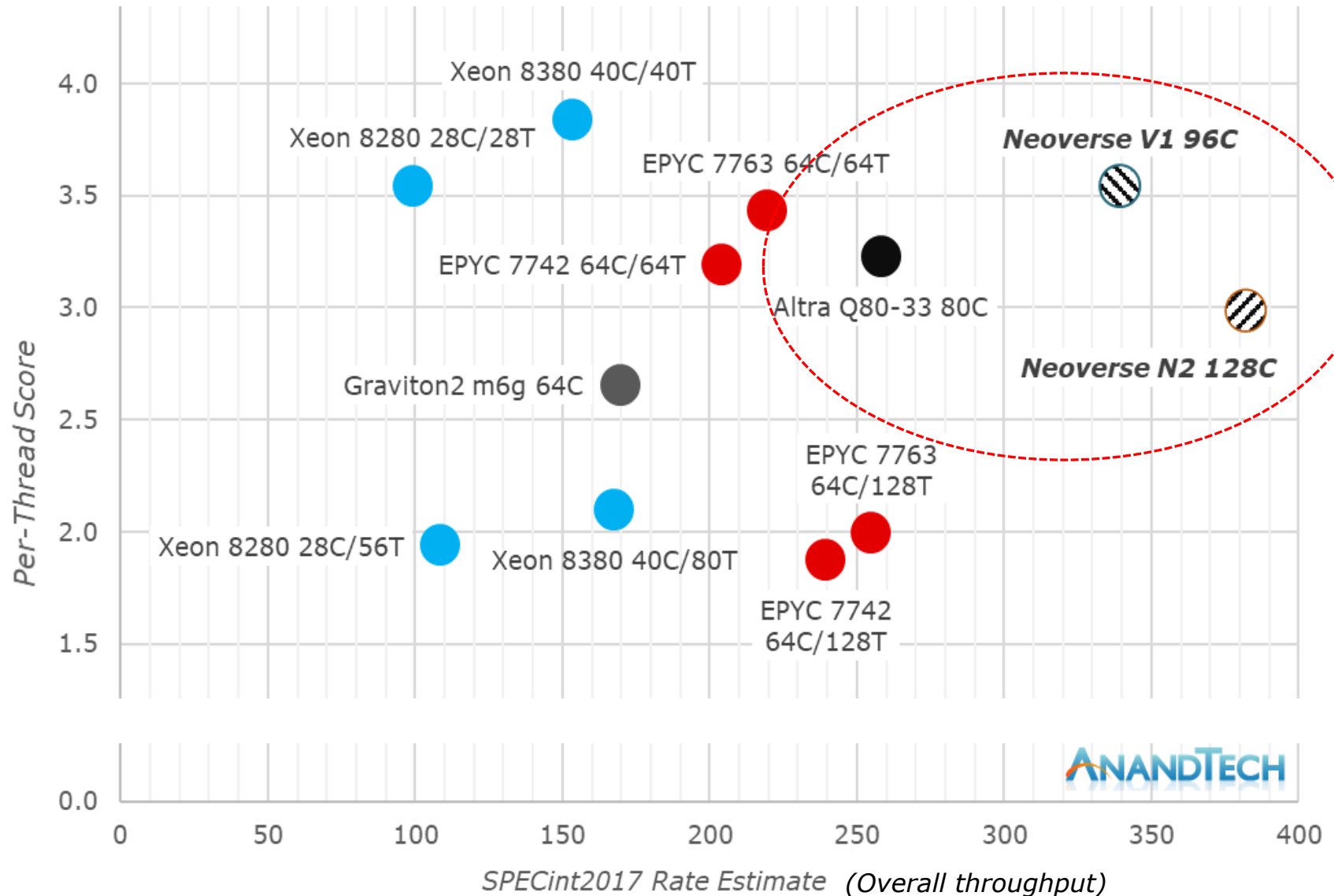
Comparing benchmark results of Intel's, Apple's, and Qualcomm's highest-performing laptop processors (as of 02/2026) [211] -1

- Based on the published benchmark results the following statements can be made:
 - When it comes to single-core Cinebench 2024 benchmarking, Qualcomm's Snapdragon X2 Elite processor surpasses Intel's Panther Lake processor, but it falls short compared to Apple's M5.
 - Due to its high core count of 18, the Snapdragon X2 Elite outperforms both Intel's 16-core Panther Lake and Apple's 10-core M5 in multi-core computing-intensive applications, as demonstrated by all other benchmarks.
 - Nevertheless, in gaming, Intel's Panther Lake significantly outperforms Qualcomm's Snapdragon X2, according to other sources [212].
 - It is important to take these results with a grain of caution, as Qualcomm's Snapdragon X2 processors have not yet been released to the market, and reliable data on the TDP of this processor and the battery life of laptops based on Qualcomm's new processor. is not available.
- Given the relatively low market share of ARM-based Windows processors, which is about 10 % [213], there are currently no expected consequences for Intel's market position following the release of Qualcomm's Snapdragon X2 family.

2.2 The race for performance leadership (26)

b) The performance competition of server processors as of 2021 -1

Let's compare per-thread and SPECint2017 rate performance scores of competing server processors from Intel, AMD and Ampere (ARM-based) [16]



2.2 The race for performance leadership (27)

The performance competition of server processors as of 2021 -2

- As shown in the Figure, Intel's 3rd generation scalable Xeon 8380 led in single-threaded performance in 2021, while AMD's Zen 3-based EPYC 7763 delivered the highest throughput for parallel workloads.
- However, this ranking is subject to change with the release of newer, more advanced server processors from Intel or AMD.
- It is worth noting that the latest ARM ISA-based server processors now deliver performance comparable to current x86-based processors from Intel and AMD, while typically consuming less power and being less expensive.
- The lower power consumption of ARM-based server processors is largely due to the inherent advantage of their RISC-type ARM ISA compared to CISC-type x86 ISA-based designs.

This is because RISC processors, such as ARM-based designs, utilize fixed-length instructions and rely exclusively on register operands (excluding LS instructions), which reduces the power required for decoding instructions and fetching operands.

In contrast, CISC processors, like those based on the x86 ISA, must handle variable-length instructions and frequently fetch operands from memory, resulting in higher power consumption.
- According to industry sources, in 2005 ARM-based servers represent approximately 20 % of total server units shipped worldwide.

2.3 The race for efficiency leadership in laptop processors

2.3 The race for efficiency leadership in laptop processors

- In 06/2024, Qualcomm unveiled its new Snapdragon X processor line, built on the ARM ISA (specifically, the Armv8.7 version) and run on Windows 11.
With the introduction of these processors, a new era in the pursuit of efficiency leadership has begun.
- These processors are manufactured using advanced 4 nm process technology and include three core clusters, each containing four high-performance Onyx cores. Designed to power laptops running Windows 11 version 24H2.
- Qualcomm claims that these processors deliver superior performance while offering significantly greater energy efficiency – resulting in longer battery life compared to existing x86-based processors from Intel and AMD.
- These claims are illustrated in the figures that follow.

2.3 The race for efficiency leadership in laptop processors (2)

Cinebench 2024 SC benchmarks of recent Windows laptops (of 08/2024) [48]

Apple M3 141 Points +15%
Apple MacBook Air 13 M3 8C GPU

Apple M2 Pro 123 Points 0%
Apple MacBook Pro 14 2023 M2 Pro

Qualcomm Snapdragon X Elite X1E-80-100 123 Points 0%
Microsoft Surface Pro OLED Copilot+

Qualcomm Snapdragon X Plus X1P-64-100 109 Points -11%
Microsoft Surface Pro Copilot+

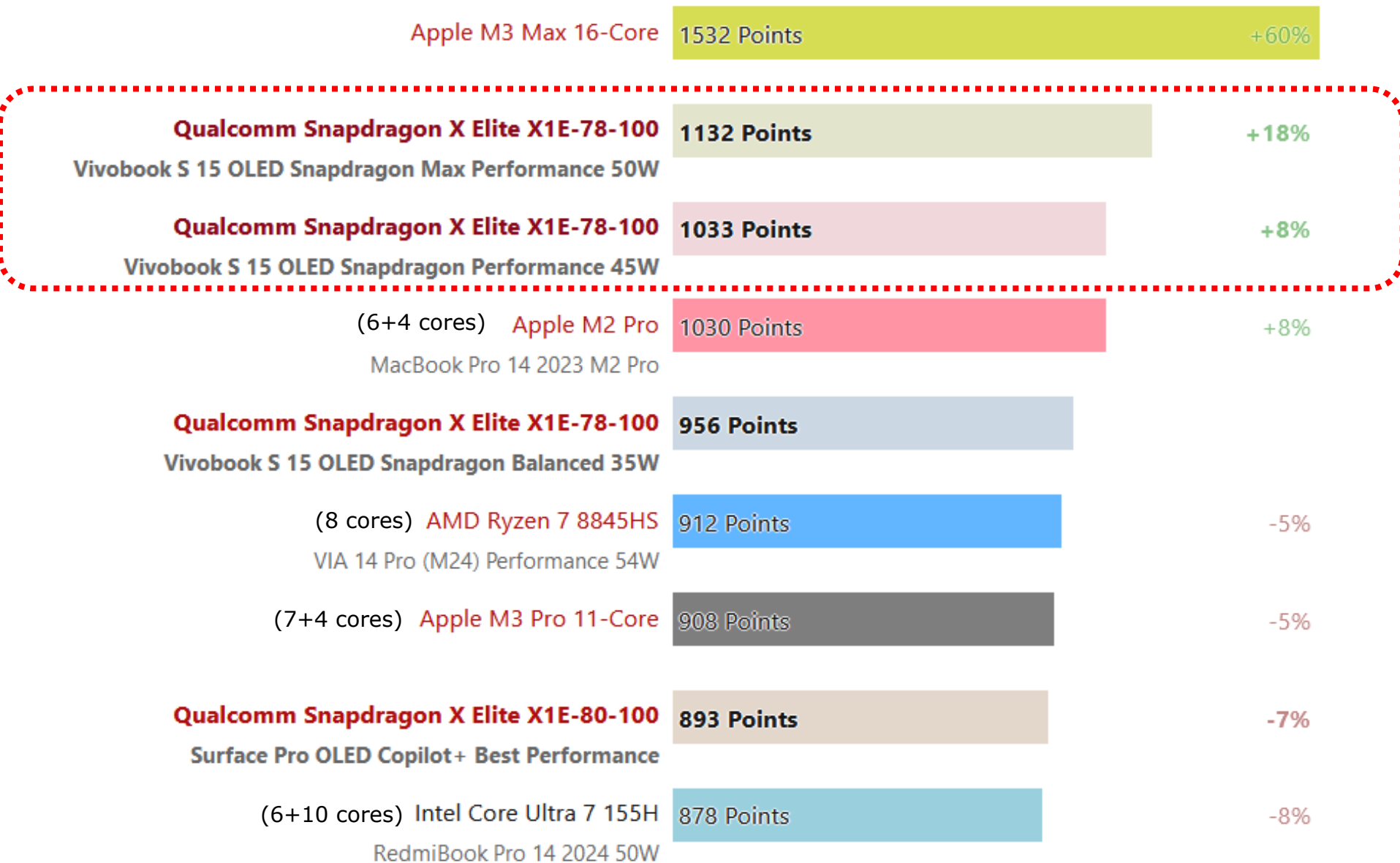
Qualcomm Snapdragon X Elite X1E-78-100 108 Points -12%
Asus Vivobook S 15 OLED Snapdragon

(Zen 4-based) **AMD Ryzen 7 8845HS** 102 Points -17%
SCHENKER VIA 14 Pro (M24)

(14th gen. Meteor Lake-based) **Intel Core Ultra 7 155H** 102 Points -17%
Xiaomi RedmiBook Pro 14 2024

2.3 The race for efficiency leadership in laptop processors (3)

Cinebench 2024 MC benchmarks of recent Windows laptops (of 08/2024) [48]



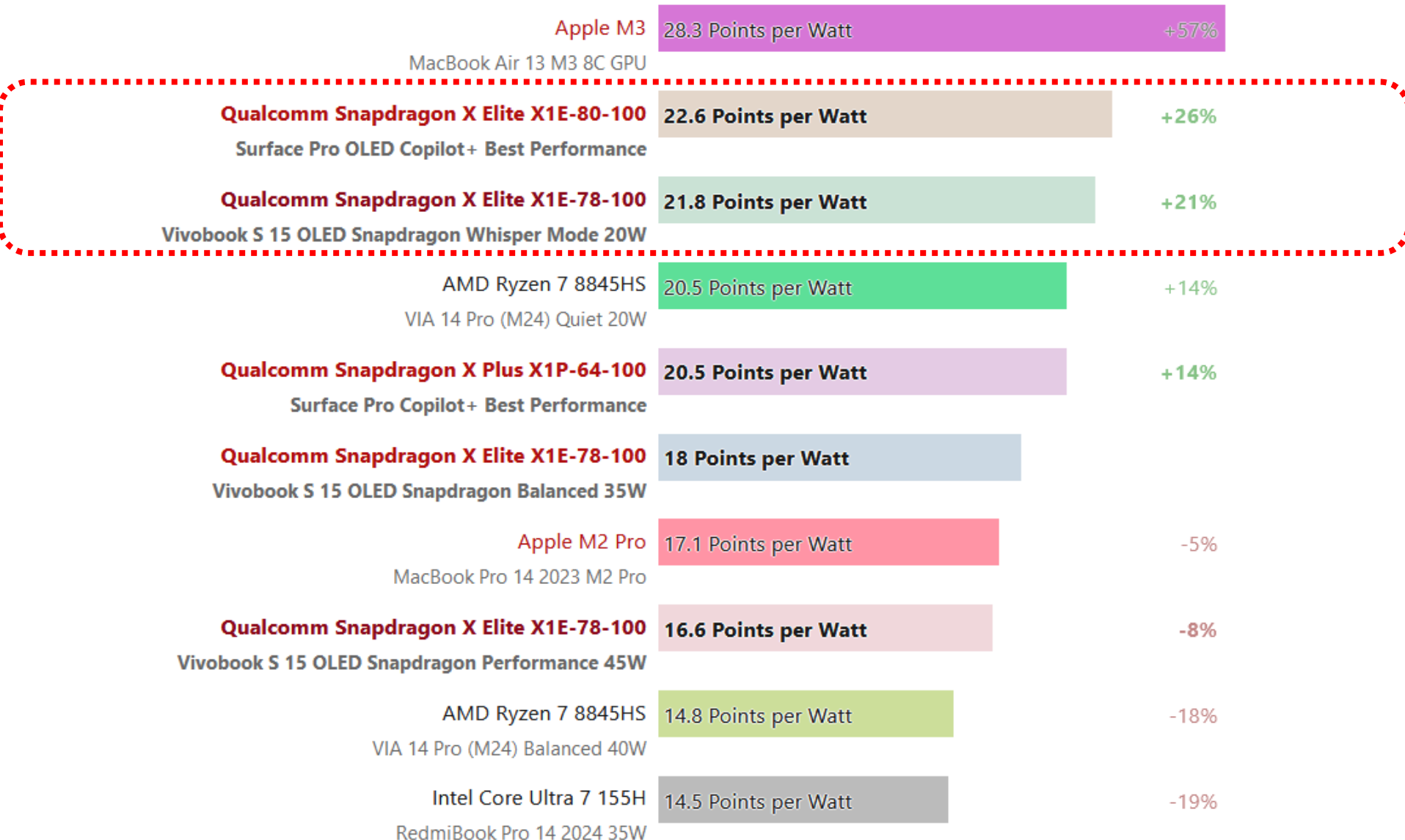
2.3 The race for efficiency leadership in laptop processors (4)

Performance comparison of recent Windows laptops

- Preceding graphs demonstrate an **astonishing finding**: **Windows laptops powered by Snapdragons X processors consistently overperform those equipped with Intel's and AMD's competing counterparts.**
- The advantage becomes even more pronounced when evaluating **efficiency**: Snapdragon X-based devices **deliver higher Cinebench 24 benchmark scores per watt of power consumed** (excluding display power), underscoring their superior performance-per-watt ratio.
- These results will be shown next.

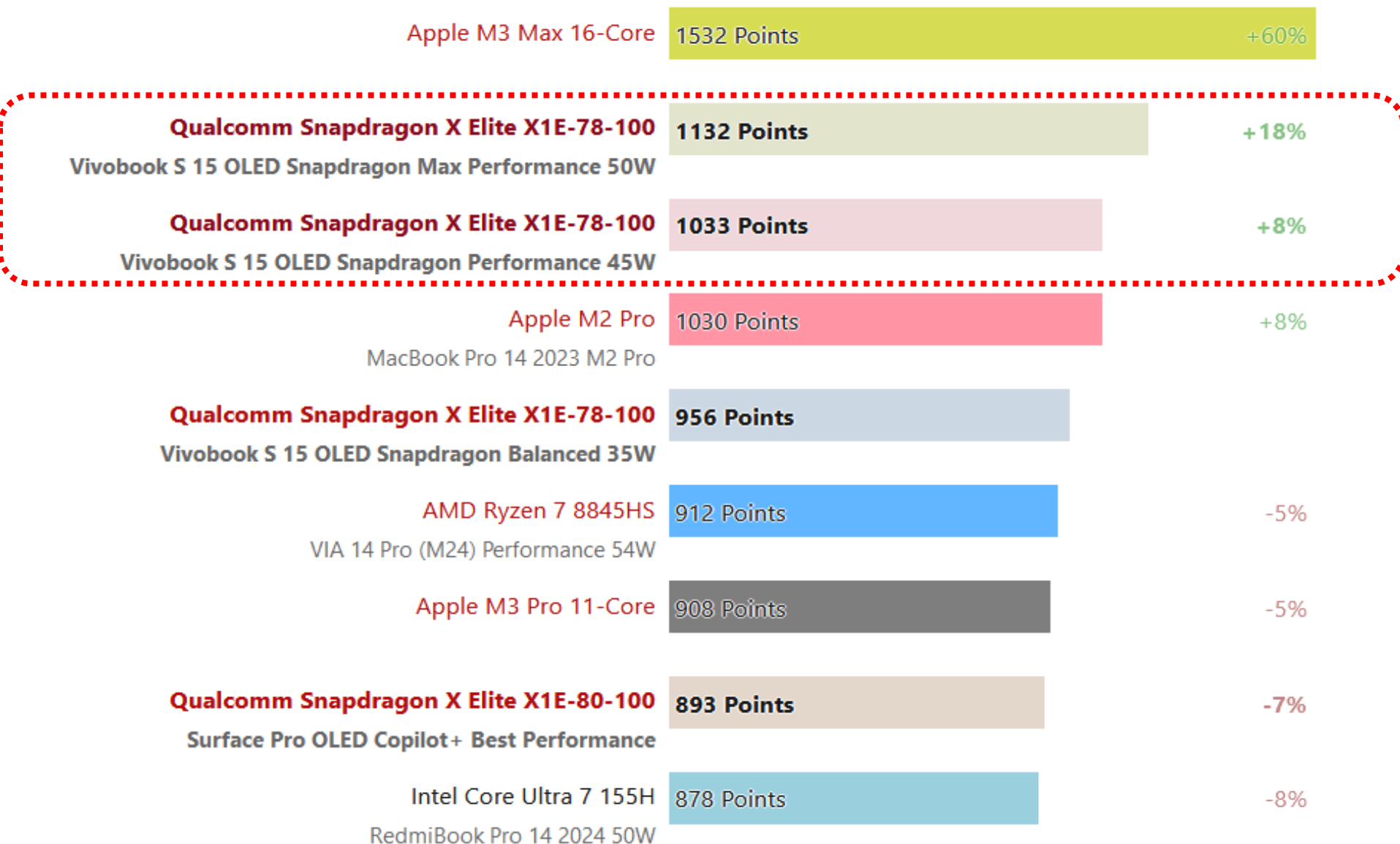
2.3 The race for efficiency leadership in laptop processors (5)

The efficiency of recent laptops, as of 08/2024 (while using external monitors) [48]
(in terms of Cinebench 2024 SC benchmark scores/Watt power consumption)



2.3 The race for efficiency leadership in laptop processors (6)

The efficiency of recent laptops, as of 08/2024 (while using external monitors) [48]
(in terms of Cinebench 2024 MC benchmark scores/Watt power consumption)



2.3 The race for efficiency leadership in laptop processors (7)

Comparing the energy efficiency, which impacts battery life in recent Windows laptops

The preceding Figures clearly demonstrate that [ARM-based Windows laptops outperform their x86-based counterparts](#) in terms of energy efficiency, resulting in superior battery life.

2.3 The race for efficiency leadership in laptop processors (8)

Recent limitations of ARM-based Windows laptops

- Despite their impressive performance and energy efficiency, ARM-based Windows laptops face a **notable drawback**: **limited application availability**.
- On one hand, the **number of native** (i.e., ARM-based) **applications** available for Windows on ARM devices **is steadily growing, but it still falls short** in comparison to the vast number of existing x86 applications developed over decades.
- On the other hand, **x86 applications can be run through emulation**.

Windows 11 24H2 introduces an advanced emulator called **Prism**, designed to improve compatibility and performance, yet emulated applications **often run noticeably slower and may suffer from compatibility issues**, particularly with certain games.

- In summary, while **ARM-based processors offer compelling advantages in energy efficiency and battery life** compared to x86-based counterparts, they **currently face significant limitations in software support**.
- Nevertheless, these challenges are expected to diminish over time.
- As developers increasingly optimize software for ARM architecture and emulation technologies continue to improve, **ARM-based laptops may well become the dominant choice for users looking for power-efficient, high-performance devices**.

2.3 The race for efficiency leadership in laptop processors (9)

Intel's first response to Qualcomm's highly efficient Snapdragon X line

- To compete with Qualcomm's Snapdragon X line, Intel has designed the highly efficient Core Ultra Series 2 processor line, known as **Lunar Lake**, for laptops.
- This series has a minimalist core configuration with **just 4 cores in total: 2 P and 2E cores**, and **it does not support multithreading** to reduce power consumption.
- A noteworthy feature of the Lunar Lake processors is the **inclusion of 16 or 32 GB in-package integrated DRAM**, which significantly reduces memory access latency.
- In addition, Lunar Lake processors integrate a **powerful NPU** unit.

This NPU plays a key role in optimizing power management by offloading certain tasks from the CPU and GPU to the more power-efficient NPU.

- Benchmark figures indicate [140] that the Lunar Lake series delivers **impressive AI performance**, along with competitive results in gaming or everyday tasks, such as web browsing, watching videos, and managing e-mails.
- However, while the Lunar Lake processors come close, **they do not fully match the energy efficiency of Qualcomm's Snapdragon X series**, as illustrated in the following Table.
- A notable advantage of the x86-based **Lunar Lake processors** is that **they offer full compatibility with x86 applications**, eliminating the emulation-related performance and compatibility issues that ARM-based Windows laptops may encounter.

2.3 The race for efficiency leadership in laptop processors (10)

Comparing NPU benchmark figures and run-down times of recent laptops, as of 12/2024 [140],

	Laptop-	Run-down time (HH:MM)	NPU TOPS	Battery
Core Ultra 7 256V Core Ultra 7 258V Core Ultra 7 256V	(Snapdragon X Plus) Dell Inspiron 14 Plus	18:19	45	54 W
	(Snapdragon X Elite) Dell XPS 13	19: 26	45	55 W
	(Intel) Dell XPS 13	17:29	45	55 W
	(Intel) Asus ExpertBook P5	14:21	45	63 W
	(Intel) Asus Zenbook S14	13:08	45	75 W
	(AMD) Asus Zenbook S16	11:35	45	78 W
	(AMD) HP Omnibook Ultra 14	12:49	50	68 W
	(AMD) Asus ROG Zephyrus G16	9:14	50	90 W
	(M3) MacBook Air 13-inch	15:13	35	52.6 W
	(M2 Pro) MacBook Pro 14-inch	17:25	15.8	70 W
	(M2) MacBook Air 15-inch	15:03	15.8	62.5 W
	(M2 Pro) MacBook Pro 16-inch	18:05	15.8	100 W

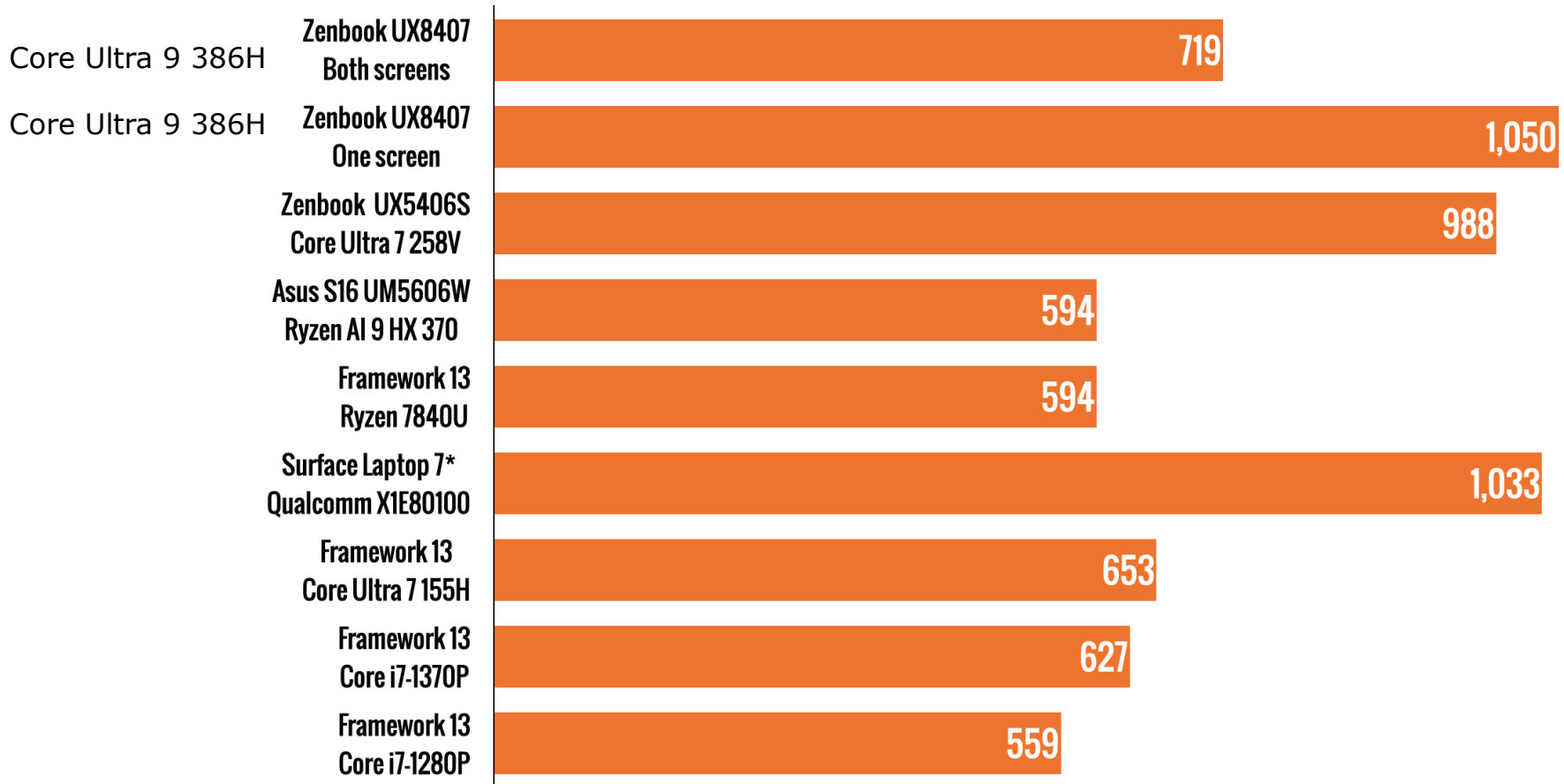
Here, we note that [Intel's processors](#) belong to the [Lunar Lake family](#) (also known as the Core Ultra Series 2), while [AMD's processors](#) are part of the [Zen 5-based Ryzen AI 300 series](#).

Intel's recent response to Qualcomm's highly efficient Snapdragon X line

- In 01/2026, Intel released its Core Ultra Series 2 (known as Panther Lake) line.
- The CPU of this line is fabricated using Intel's most advanced A18 process.
- This line not only offers exceptional performance, as previously discussed, but also demonstrates impressive power efficiency, as shown in the following Figure.

2.3 The race for efficiency leadership in laptop processors (12)

Battery life comparison using the PCMark Modern Office benchmark [207]
(Results in minutes, higher is better) [207] -1



Remark

PCMark measures how well a laptop handles light-to-medium office tasks, such as writing, web browsing, and video conferencing. It is part of the PCMark 10 benchmark (UL Benchmarks).

2.3 The race for efficiency leadership in laptop processors (13)

Battery life comparison using the PCMark Modern Office benchmark [207]
(Results in minutes, higher is better) [207] -2

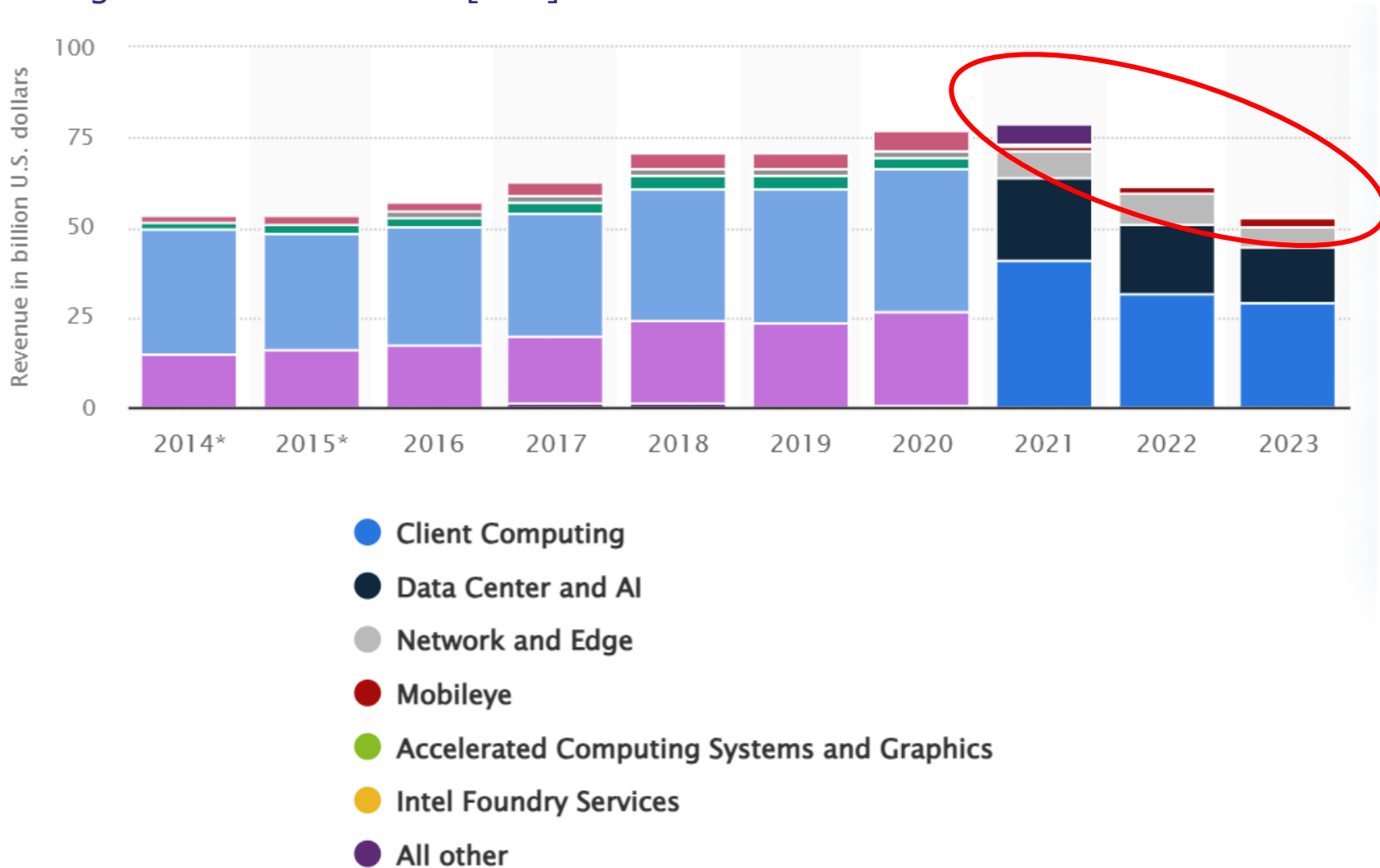
- The Figure above clearly shows that Intel's Core Ultra Series 3 processors (known as Panther Lake processors) offer outstanding power efficiency, resulting in longer battery life.
- However, this comparison pertains to earlier processors, which were manufactured using less advanced IC technology and, as a consequence, exhibit higher power consumption.
- This situation may change once competing processors using more advanced manufacturing processes will hit the market.

2.4 What is Intel's current market position?

3. What is Intel's current market position? (1)

3. What is Intel's current market position?

From 2021 to 2023 Intel's revenues have decreased across most market segments, as the Figure below illustrates [137]



3. What is Intel's current market position? (2)

Consequences of Intel's declining revenues -1

a) Share prices have been declining from 2021 to 2025.



Intel's share price dropped from over 60 \$ in 2021 to around 20 \$ by mid-2025, losing approximately 2/3 of its value over that period.

3. What is Intel's current market position? (3)

Consequences of Intel's declining revenues -2

- b) Pat Gelsinger, Intel's CEO since 2021, announced his resignation in 08/2024 effective 12/2024.
- c) In 08/2024, Intel unveiled plans to cut approximately 15000 jobs, about 15 % of its workforce, by year-end, aiming to save \$10 billion.
- d) Intel's current CEO is Lip-Bu Tan, who was appointed in 03/2025.
Born in Malaysia and raised in Singapore, he is a former CEO of Cadence Design System.

3. What is Intel's current market position? (4)

Latest developments in Intel's market position

- In 08/2025, the United States government announced a landmark investment of \$8.9 billion in Intel's common stock, signaling strong federal support for domestic semiconductor manufacturing.
- In 09/2025, NVIDIA acquired Intel's stock for approximately \$5 billion.
- By 01/2026, Intel began mass production using its A18 process, and launched the Core Ultra Series 3 processors (known as Panther Lake) processors. These processors outperformed competing products in both performance and power efficiency.
- Intel's stock price reflected these events, as illustrated in the following Figure.

3. What is Intel's current market position? (5)

How Intel's stock price responded to market events from Aug. 2025 to Febr. 2026)



10 b USD NVIDIA buys
from the Intel stock
government (5 b USD)

Launch of
Panther Lake

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